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Generalizing Privacy Risk Detection Models Across Social Media Platforms Through Cross-Dataset Evaluation

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Abstract

In recent years, social networks have become a central part of daily life, changing the way people communicate, share information, and present themselves online. Posts, comments, photos, and other forms of user-generated content often seem harmless, yet they can reveal much more than users realize. Personal details are exposed not only through explicit identifiers such as names, phone numbers, or addresses, but also through habits, language, context, and repeated patterns of sharing. For this reason, privacy on social media is no longer just a legal or theoretical issue, but also a practical and technological challenge.

This growing exposure has made the automatic detection of privacy-sensitive content an important area of research. Identifying risky disclosures early can promote safer online behavior and contribute to the development of systems that are more aware of how personal information is shared in digital environments. This thesis focuses on the problem of detecting privacy risk in social media content, with particular attention to how well detection models can generalize across different platforms and datasets. Although many existing approaches achieve good results within the specific dataset on which they are trained, their performance often drops when they are applied to data coming from a different source. This happens because each platform has its own language, conventions, user behavior, and annotation logic, making cross-dataset generalization a difficult but important goal. Starting from a broader reflection on privacy, digital identity, and the role of personally identifiable information in online environments, this work then moves to the computational side of the problem. Several datasets related to privacy disclosure and privacy-sensitive content are analyzed

and compared to highlight their differences in structure, labeling, and type of information.

On this basis, the thesis evaluates multiple text representations and classification models, ranging from more traditional approaches to transformer-based methods, to understand which solutions are more robust when transferred across heterogeneous contexts. A central part of the study is the cross-dataset evaluation, extended through experiments on the SynthPAI dataset. Different settings are explored, including zero-shot evaluation, decision-threshold adjustment, fine-tuning strategies, and integration of demographic metadata. This makes it possible not only to compare model but also to better understand which factors actually improve adaptation to new and unseen data. In general, this thesis shows that privacy risk detection cannot be treated as a simple classification task detached from the context. The way people disclose personal information online is complex, variable, and deeply influenced by the platform and the form of expression. For this reason, building effective models requires not only strong language representations, but also evaluation strategies that test their ability to generalize beyond a single dataset. In this sense, the work aims to contribute to a more realistic understanding of privacy detection on social media and to support the development of more reliable and transferable methods.

Chapter 1

Introduction

In contemporary digital environments, social media platforms have become central spaces for communication, self-presentation, and information exchange. Users routinely share opinions, personal experiences, images, locations, and biographical details across heterogeneous platforms such as Reddit, X/Twitter, forums, and other online communities.

Although this continuous production of user-generated content supports interaction and visibility, it also increases the likelihood that personal or sensitive information may be disclosed, inferred, or recontextualized beyond the original intentions of the user. For this reason, privacy protection in social media is no longer limited to the removal of explicit identifiers, but increasingly concerns the detection of subtle textual cues that may reveal sensitive attributes, personal habits, or vulnerable aspects of an individual’s identity.

This problem is particularly relevant in the context of automatic detection of privacy risk. In recent years, several datasets and machine learning approaches have been proposed to identify privacy-sensitive content in social media posts. However, most existing studies evaluate models within a single dataset, under platform-specific annotation schemes and domain-dependent linguistic conventions. As a consequence, strong performance on a benchmark does not necessarily imply that the same model can generalize to different datasets or social

media environments. This limitation is especially important for privacy-related tasks, where the notion of sensitivity can vary between contexts, communities, annotation guidelines, and content modalities.

Based on these considerations, this thesis investigates the generalization of privacy risk detection models across social media platforms through a cross-dataset evaluation framework. The main objective is to understand whether models trained on one dataset are able to capture transferable privacy-related signals, or whether their performance is mainly driven by dataset-specific patterns. In other words, the thesis examines to what extent privacy-sensitive content detection can be treated as a robust cross-domain task rather than as a collection of isolated benchmark problems.

To address this research question, the thesis first introduces the broader conceptual background of privacy and digital identity in online ecosystems. Then it reviews a set of datasets proposed in the literature for privacy-related content analysis, highlighting their differences in source platform, annotation methodology, target phenomena, and level of granularity. On this basis, the work defines a common experimental framework for comparing heterogeneous resources in a unified methodological setting.

The empirical part of the thesis focuses on the construction of a cross-dataset evaluation pipeline. The selected datasets are analyzed, harmonized and reformulated in order to support a common classification setting. The different textual representations and machine learning models are then considered to assess performance both within individual datasets and across different datasets. This design makes it possible not only to evaluate predictive effectiveness, but also to measure the extent to which the learned representations remain stable when transferred to data collected in different social media contexts.

The contribution of the thesis is therefore twofold. First, it provides a comparative perspective on privacy-related datasets by organizing them within a shared analytical framework. Second, it experimentally investigates the generalization limits of privacy risk detection models, paying particular attention to the gap between in-domain performance and cross-dataset transferability. More

broadly, the work contributes to the discussion on privacy-aware machine learning by showing that the practical usefulness of a detector depends not only on accuracy within a benchmark but also on its robustness across heterogeneous settings.

The remainder of the thesis is organized as follows. Chapter 2 presents the conceptual background on privacy, digital identity, and the role of social media in personal information disclosure. Chapter 3 reviews the main datasets and studies related to privacy-sensitive content detection and cross-platform privacy analysis. Chapter 4 describes the methodological framework adopted in the thesis, including data preparation, feature extraction and the model design. Chapter 5 reports the experimental results and discusses the main findings emerging from both within-dataset and cross-dataset evaluation. Finally, Chapter 6 summarizes the main conclusions of the work and outlines possible future research directions.

Chapter 2

Privacy and Digital Identity in the Social Media Ecosystem

This chapter provides the conceptual background necessary to frame the problem addressed in this thesis. In particular, it introduces the main dimensions of privacy in digital environments, the role of personal data disclosure in social media, the relevance of Personally Identifiable Information (PII) in the construction of digital identity, and the privacy implications of unstructured user-generated content.

2.1 Introduction

Digital transformation has fundamentally altered the way in which personal data is perceived and managed. The widespread adoption of the Internet and social media platforms has transformed every individual into a perpetual producer of information, much of which is shared unconsciously and remains difficult to control.

This phenomenon extends far beyond 'traditional' data such as names, addresses, or phone numbers to encompass digital interactions, user-generated content, biometric signals, metadata, and behavioral traces that, despite appearing innocuous, have become valuable assets for profiling, surveillance, and economic exploitation.

This chapter addresses four fundamental aspects of this scenario:

- the evolution of privacy as a concept and the various definitions that have shaped its understanding;
- the unconscious dissemination of data by users on the web and social media, including associated risks and beneficiaries;
- the role of Personally Identifiable Information (PII) in constructing digital identity and governing data;
- the privacy implications of unstructured and user-generated content, including photos, audio files, and social media posts.

2.2 Privacy: Definitions and Evolution of the Concept

The notion of privacy has undergone significant conceptual evolution since its earliest formulations, reflecting technological, social, and legal transformations in society. Initially conceived as the right to “be left alone” in Warren and Brandeis’s famous late-nineteenth-century definition, privacy has progressively evolved into a polysemic concept encompassing individual, relational, and collective dimensions.

In the European context, the Charter of Fundamental Rights of the European Union distinguishes between the right to respect for private life (Article 7) and the right to protection of personal data (Article 8), signalling a departure from the traditional conception of privacy as merely a sphere of intimacy to be

shielded from others’ scrutiny. As Rupp and von Grafenstein observe, the protection of personal data should not be understood as the protection of a “good” in itself, but rather as an instrument for safeguarding the autonomous exercise of fundamental human rights threatened by data processing in increasingly pervasive digital contexts [1]. From this perspective, privacy evolves from a static barrier into a dynamic mechanism for managing information-related risks.

Attention shifts from focusing solely on the content of data to examining the context of processing, the purposes pursued, and the effects produced. This marks the transition from a proprietary conception of privacy to a relational and situated vision that measures harm as a function of the probability of impact on fundamental rights. This approach is reflected in European positive law. While the GDPR does not expressly define privacy, it embodies an evolved understanding through principles such as transparency, minimization, purpose limitation, and accountability, which together configure a comprehensive model of “data governance”. The EDPB has emphasised that these principles must be implemented concretely in the design of digital interfaces and services (privacy by design and by default), demonstrating that privacy is no longer merely a defensive right but a technical-legal criterion that must guide system architecture [2]. Within this framework, privacy itself assumes an increasingly procedural rather than substantialist character: not a predetermined inviolable domain, but the result of continuous balancing between individual freedoms, collective needs, and technological innovation.

A significant aspect of this transformation is the expansion of the notion of sensitive data. Quinn and Malgieri demonstrate how technological evolution – particularly advances in computational capacity and correlation techniques – has made it increasingly difficult to maintain a clear distinction between “sensitive” and “non-sensitive” data, since even apparently neutral information can, when combined, reveal deeply intimate traits of individuals [3]. Privacy is therefore no longer reducible to the protection of certain data categories but has become a system of safeguarding against discriminatory and harmful uses of information per se. In this sense, jurisprudence from the European Court of Human Rights, as referenced by the authors, confirms that the protection of sensitive data

is an “end in itself”, directly connected to human dignity rather than merely a means to prevent specific forms of discrimination. This demonstrates how privacy, originally understood as a space of confidentiality, has evolved into the broader notion of “informational self-determination” – the right of individuals to maintain effective control over information concerning them, despite the growing difficulty of exercising this right in a digital ecosystem dominated by information asymmetries and economic models based on profiling.

The evolution of the concept of privacy thus reveals a dual movement. On one hand, protection has expanded beyond the intimate sphere to encompass the public and collective space of digital interactions. On the other hand, privacy has transformed into a structural principle of the legal order, capable of guiding both normative choices and the technical and organizational solutions adopted by data controllers. Ultimately, contemporary privacy is no longer simply the right to hide, but rather the foundational condition for guaranteeing substantive freedoms: the ability to express oneself without fear of surveillance, to make choices free from hidden manipulation, and to participate authentically in social and political life. This trajectory reflects the ongoing evolution of the concept under the combined pressures of technological transformation and legal scholarship, posing new challenges to legislators, supervisory authorities, and individuals themselves as they exercise their digital citizenship.

2.3 The Web and Social Media: User Unawareness, Risks, and Hidden Beneficiaries of Data

In contemporary digital society, the massive use of the web and social media has radically transformed the relationship between individuals and information, generating new dynamics in the production and circulation of personal data. Users often unconsciously disseminate enormous quantities of information about themselves daily through seemingly trivial actions: clicking “like”, sharing a photo, checking a route, or hastily accepting a consent request presented by an interface. What appears to be an ordinary gesture of participation in online

social life serves as the starting point for a complex process of data collection, aggregation, and analysis that, while fuelling the business models of digital platforms, poses serious risks to fundamental rights.

The central issue is that the distinction between personal and non-personal data is not always clear. As Rupp and von Grafenstein highlight, apparently neutral data, such as technical or geolocation data, become “personal” in relation to the context in which they are processed and the purposes for which they are used, since their processing can generate specific risks for individuals’ fundamental rights [1]. The authors propose an analytical framework based on three elements: “about, purpose, and result”. This framework demonstrates how risk derives not only from the content of data but also from the intended purpose and the outcome of processing. Anonymous data can be reconnected to an individual or used to evaluate their behaviours and preferences, producing concrete effects on their autonomy and life opportunities.

This perspective is crucial for understanding how users, believing they are revealing nothing significant, inadvertently expose intimate aspects of their identity that can be exploited by third parties. An additional element of criticality, emphasised in the European Data Protection Board’s Guidelines 03/2022, concerns so-called deceptive design patterns: many social platforms implement interfaces and user pathways that, far from being neutral, deliberately orient users’ decisions toward choices favourable to the platform’s interests and contrary to the user’s own, particularly regarding data management [2]. Techniques such as “overloading”, which overwhelms users with excessive requests or options; the “privacy maze”, which makes protection settings difficult to locate; “stirring” which manipulates choices by exploiting emotions; or “left in the dark” which obscures relevant information, induce users to share more data than necessary or to consent to processing they do not fully understand. This represents not merely individual inattention but genuine exploitation of cognitive biases and psychological vulnerabilities that drastically reduce the possibility of free and informed consent, in direct contravention of the principles of fairness, transparency, and accountability mandated by the GDPR. The outcome is that masses of data flow toward platforms without users having full awareness of the

associated risks – risks that become even more serious when the processed data assumes a “sensitive” nature.

On this point, Quinn and Malgieri have demonstrated how the notion of sensitive data, traditionally limited to well-defined categories (ethnic origin, religious beliefs, health, sexual orientation, political opinions, and trade union membership), is now in continuous expansion. The increase in computational power and correlation techniques enables the extraction of sensitive information even from apparently innocuous data [3]. For example, online purchase histories, shared photographs, and social media interactions can, once processed with predictive algorithms, reveal a person’s health status, political preferences, or sexual orientation, exposing them to discrimination or differential treatment. The difficulty of defining and delimiting the category of sensitive data entails the risk of “conceptual inflation”, a progressive erosion of the distinction between ordinary and special categories of data. The consequence is that an increasing volume of data can, in practice, be qualified as deserving enhanced protection.

This scenario accentuates the dangers for the most vulnerable populations: minors, elderly individuals, and those with limited digital literacy, who not only fail to understand the implications of their online actions but are also more susceptible to mechanisms of hidden influence. The primary beneficiaries are large digital platforms that base their economic models on profiling and targeted advertising, transforming users’ attention and data into commodities in the digital economy. Advertisers leverage such information as a powerful tool for influencing consumption patterns, exploiting cognitive and emotional vulnerabilities. Political actors and governments can manipulate opinions and electoral behaviours, as demonstrated by the Cambridge Analytica scandal. Criminal organizations and malicious actors can exploit digital traces for fraudulent or persecutory purposes. In this ecosystem, the informational power asymmetry between users and platforms becomes profound: while the former navigates an apparently free and benign environment, the latter accumulates knowledge and decision-making power based on data that users have provided without full awareness. This reveals a problematic framework in which fundamental rights risk being compromised not only through illicit processing but also through

formally legitimate practices that nonetheless fall within regulatory grey areas. The web and social media therefore represent an ambivalent space: on the one hand, vehicles for freedom of expression and democratic participation; on the other, instruments of systematic information collection whose economic and political valorisation often occurs at the expense of individuals’ dignity, autonomy, and non-discrimination.

This scenario demands greater user awareness, strengthened platform transparency and accountability, and the development of regulatory strategies capable of balancing innovation with effective protection of fundamental rights. Only through such measures can the gap be reduced between the perception of online freedom and the reality of an environment increasingly characterised by hidden exploitation of personal data – an environment in which risks for individuals grow proportionally to the economic and political benefits accrued by those who collect, process, and monetise that data.

2.4 The Role of Personal Identifiable Information in the Digital Ecosystem: Identity, Correlation, and Tools of Invisible Control

In the contemporary digital society, the concept of Personal Identifiable Information (PII) plays a central role in understanding how individual identity is progressively tracked, reconstructed, and exploited. Although European law does not use the term PII – preferring instead “personal data” – this concept, widely used in the United States and technical computing fields, provides a useful framework for understanding the pervasive nature of information collection in the digital era. According to the U.S. Department of Labor, PII consists of “information that can be used to distinguish or trace an individual’s identity, either alone or when combined with other information that is linked or linkable to a specific individual” [4]. This definition highlights two fundamental elements: first, the capacity to distinguish or trace an individual; second, the potential for correlation with other data to achieve identification.

In the technical domain, PII is further divided into linked PII, data that directly identify individuals, such as name or social security number, and linkable PII, information that, taken individually, does not identify a person but becomes identifying when combined with additional datasets [5]. This second characteristic is particularly concerning, as it transforms apparently innocuous data (such as browsing preferences, geolocation metadata, or device characteristics) into pieces of an identity mosaic. Within this logic, PII represents not only information that reveals who we are, but also a collection of weak signals that, through data fusion and entity resolution techniques, enable the reconstruction of unique and persistent personal profile [6].

The European framework approaches PII differently. Although the term itself is not expressly used in the GDPR, its substance is largely encompassed within the European definition of personal data. Article 4(1) of the GDPR defines personal data as “any information relating to an identified or identifiable natural person”, specifying that an individual is identifiable even indirectly, through reference to elements characteristic of their physical, physiological, genetic, mental, economic, cultural, or social identity [7]. This confers upon “personal data” a scope sufficiently broad to implicitly include all variants of PII. However, while the European perspective approaches data protection as a safeguard of fundamental rights, the United States approach tends to treat PII more pragmatically, focusing on security, the risk of identity theft, and the prevention of economic abuses [8].

This distinction between regulatory frameworks underscores a particularly relevant technical issue: the difference between direct identifiers and quasi-identifiers. Direct identifiers (such as tax identification numbers, passport numbers, or residential addresses) permit immediate identification of the subject. Quasi-identifiers (such as date of birth, postal code, or gender) appear neutral individually but can become highly identifying when combined. Statistical research has demonstrated that with only a few quasi-identifiers, it is possible to re-identify the majority of individuals in a dataset, thereby undermining the promise of anonymity [9]. In this sense, PII illuminates the problematic nature of de-identification and pseudonymisation techniques, which often fail

when confronted with computational power and the increasing availability of ancillary data.

The significance of PII extends beyond the technical dimension to directly implicate the digital economy. Identifying data represents the key to connecting dispersed fragments of information and transforming them into unique and stable profiles. They constitute the raw material for the business models of large platforms, which are based on profiling, behavioural advertising, and microtargeting [10]. The personal identifier, far from being merely a neutral attribute, becomes the element that enables the linking of every user interaction – clicks, searches, purchases, movements – to a permanent archive, thereby determining the possibility of predicting needs and desires while simultaneously shaping future behaviours. This process remains invisible to the user, who experiences the digital environment as free and neutral, while in reality constantly surrendering PII that fuels value chains concentrated in the hands of a few global actors.

Alongside its economic dimension, PII has profound political and social implications. The systematic use of personal identifiers enables not only marketing practices but also instruments of surveillance and control. In the governmental sphere, the combination of PII with public registers, tax data, or video surveillance systems permits the creation of interoperable databases that can be deployed for security and prevention purposes. In authoritarian contexts, such practices have led to forms of widespread social control. Yet even in advanced democracies, the extensive use of PII can erode spaces of freedom and anonymity, reducing the possibilities for dissent and authentic civic participation [11].

The centrality of PII has prompted several non-European legal systems to develop specific regulations. China’s Personal Information Protection Law (PIPL), for instance, defines “personal information” as “any type of information identifying or capable of identifying a natural person”, subsequently distinguishing sensitive information that, if compromised, may cause harm to an individual’s dignity and imposing more stringent obligations for such data [12]. Similarly, the California Consumer Privacy Act (CCPA) treats “personal information” as

a broad category that includes identifiers, biometric data, and financial information, mandating transparency obligations and opt-out rights for users [13]. These developments demonstrate how the notion of PII has become a cross-cutting category in global regulations, attesting to its importance for protecting individuals in the digital ecosystem.

Ultimately, the concept of Personal Identifiable Information functions as a bridge between fragmented data and unified identity. It represents the point of convergence among the technical sphere (which enables identification), the legal sphere (which falls within the perimeter of regulatory protection), and the economic-political sphere (which enables profiling, monetization, and control). As privacy evolves as a right to informational self-determination, PII constitutes the technical-legal foundation that makes identity reconstruction possible. The challenge is no longer simply to protect individual attributes, but rather to disrupt the invisible connections that link fragments of information into omnipresent personal profiles. Only through this approach will it be possible to balance innovation with the protection of fundamental rights, ensuring that the construction of digital identity remains in the hands of individuals and does not become a commodity exclusively available to those who hold the power to aggregate and exploit data.

2.5 Privacy and Unstructured Content: The Invisible Challenges of User-Generated Content

One of the most pressing challenges for privacy protection in the digital context concerns unstructured content – the body of data that does not follow predefined schemas and encompasses images, videos, audio files, free-form texts, and user-generated content (UGC). Unlike structured data, which is numerical or tabular and easily classifiable and processable, unstructured data is fluid, heterogeneous, and difficult to reduce to standardized formats [14]. Its semantic richness and massive diffusion across social platforms make it a valuable source

for algorithmic analysis, but simultaneously a growing threat to individual privacy.

According to recent estimates, over 80% of globally generated data falls within the unstructured category [15]. This means that the majority of users' daily online interactions – shared photos, social media posts, voice messages, short videos – constitute an informational resource not immediately governed by structures, yet highly exploitable. The problem arises from the fact that this content not only conveys explicit information (a text, an image) but also contains hidden metadata, such as GPS coordinates, timestamps, EXIF data from images, and technical parameters of devices. These elements, often overlooked by users, can be utilized to trace locations, habits, and movements, rendering even content that appears anonymous identifiable [16]. Digital photographs provide a paradigmatic example. A photo posted online can reveal not only the face of the depicted person, analyzable through facial recognition systems, but also the location where it was taken, the precise time, the device model used, and even environmental details that, when combined with other data, enable precise reconstruction of private life. Similarly, an audio file can be processed to extract not only the words spoken but also vocal timbre, emotional state, and even biometric characteristics. Semantic analysis of free-form text can reveal political convictions, religious orientations, or psychological traits. In this manner, unstructured content becomes an instrument of inference far more powerful than it appears to the user who generates it.

The phenomenon assumes further relevance when one considers the centrality of user-generated content in the social web. Posts, comments, images, videos, and podcasts constitute the backbone of digital communication and are often produced spontaneously, with limited awareness of the risks associated with their algorithmic reprocessing. The deployment of machine learning and deep learning techniques enables the analysis of such content to extract implicit information: emotions, moods, political preferences, consumption habits, and psychographic inclinations [17]. In several cases, content uploaded by users has been utilized to train artificial intelligence systems without the creators' awareness of the role attributed to their data [18]. This introduces a crucial

ethical question: to what extent can freely shared content be repurposed for ends different from those for which it was generated?

A particularly problematic aspect concerns the anonymization of unstructured content. While consolidated methodologies exist for structured data (suppression, generalization, noise addition), de-identification is far more complex for unstructured data. As Weitzenböck observes, rendering free-form texts or images truly anonymous is nearly impossible: content can always be reconnected to an individual through correlations with other information sources [19]. For instance, a textual post mentioning frequented locations can be cross-referenced with location data from other applications; an image of an urban landscape can be recognized by image-matching algorithms and linked to a precise address. This difficulty is compounded by the semantic heterogeneity of unstructured content. Such content conveys implicit meanings, metaphors, and cultural references that elude user control and are reinterpreted by algorithms in unpredictable ways. Machine learning models, often operating as “black boxes”, extract patterns invisible to the content producer, thereby amplifying informational asymmetry. Users know neither what algorithms can deduce from their posts nor how these inferences are subsequently utilized for commercial or surveillance purposes.

The issue also intersects with platform responsibility. Hosting providers that host user-generated content occupy an ambiguous position: formally neutral with respect to uploaded material, they in reality play an active role through indexing, moderation, and algorithmic recommendation. While European Union law tends to guarantee a limited liability regime for third-party content, the activity of analysing and exploiting unstructured data generates new transparency and protection obligations that challenge the traditional immunity model [20].

A further critical issue concerns the lifecycle of unstructured content. Photos, videos, and texts can persist in archives, backup copies, foreign servers, or snapshots even after apparent deletion by the user. This phenomenon undermines the effectiveness of the right to be forgotten: what has been shared may remain accessible or reemerge in future contexts, with consequences for personal

reputation and dignity. The asymmetry between user perception, in which users believe they have deleted content, and technical reality – in which the content persists - represents one of the most serious threats to individual informational sovereignty.

Finally, the risk of algorithmic bias and discrimination must be considered. The analysis of unstructured data can incorporate implicit stereotypes: facial recognition algorithms have demonstrated systematic errors with respect to people of colour; semantic analysis systems can associate gender or ethnic characteristics in distorted ways; speech recognition models perform more poorly with minority accents. This means that user-generated content, far from being neutral, can become an instrument of exclusion or reinforcement of social inequalities when interpreted by inequitable technological systems [21].

2.6 Related Work

This thesis builds upon the works presented in [22], [23], who proposed some models for assessing user privacy on social media. In their paper “Assessing User Privacy on Social Media: The Twitter Case Study” [22], the authors proposed a framework based on the Vector Space Model (VSM) that identifies four distinct categories of privacy-relevant features. Profile-based features model the personal data disclosed during registration (date of birth, occupation, address, profile URL); behavior-based features capture user behavior on the platform (number of likes, number of posts); network-based features represent the size and characteristics of the user’s social network (number of followers, contacts, public lists); and, crucially, content-based features analyze the textual content published by users, revealing sensitive information identified through lists of terms associated with ten categories inspired by the GDPR: health, ethnicity, religion, political orientation, sexual orientation, geolocation, profession, marital status, interests/passions, and age. The model represents each user’s privacy profile as a vector in an n -dimensional space, and the overall privacy score is calculated as the Euclidean distance between the user’s privacy vector and an ideal vector representing a completely protected user. The authors tested the

model through three distinct tasks: a binary classification (at risk/not at risk), a multi-class classification (low risk, moderate risk, intense risk, high risk), and a ranking of users by privacy risk, achieving 75% accuracy in binary classification and an NDCG (Normalized Discounted Cumulative Gain) greater than 0.94 in the ranking task on a dataset of 1,500 Twitter users.

In their next work, “Unveiling the Privacy Risk: A Trade-Off Between User Behavior and Information Propagation in Social Media” [23], they developed an unsupervised model that generates a risk score by explicitly combining two fundamental and previously disjointed dimensions: the direct release of sensitive information and the propagation of content through social networks. The model combines two distinct scores. The first, called the Sensitive Information Release Risk Score (SIRRS), assessed the tendency of the user (and other users who mention them) to release sensitive information, considering both the user’s direct tweets and tweets that mention them, their biography, and the geolocation of their profile. The second score, the Potential Scope Risk Score (PSRS), identified tweets that achieve a level of interaction (likes, retweets, comments) significantly higher than the user’s average, indicating a high potential for exposure to a wider audience than normal. Finally, the Global Privacy Risk Score (GPRS) was calculated by linearly combining SIRRS and PSRS through different weight combinations to assess the overall risk.

In addition to the unsupervised model, they proposed a supervised model based on machine learning algorithms (Logistic Regression, K-Nearest Neighbors, Random Forests) that used twenty metadata features and textual features extracted using TF-IDF (unigrams, bigrams, trigrams) for automatic risk classification. Experimental results on a dataset of 100 Twitter users manually annotated by twelve human assessors showed an accuracy of 78% for the unsupervised model in binary classification and 87% for the supervised model based on Random Forests with bi-tri-gram features, demonstrating that the integration of textual information significantly improved predictive performance. The particular novelty of this framework is how it addresses the trade-off between the individual release of sensitive information and its dissemination through social circles. The results suggested a positive interaction between these factors:

users who released more sensitive information tended to see this content amplified across the network. This aspect is particularly relevant for understanding unwitting disclosure, as users may not be fully aware of the exponential potential for propagation of their posts when the content combines sensitivity and virality.

Before the work of Livraga and Viviani [22], [23], one of the first approaches to this problem was proposed by Liu and Terzi [24]. The authors assigned a privacy score to users considering the combination of data sensitivity and visibility on the social platform, using Item Response Theory for the calculation. Liu and Terzi’s framework is based on the observation that privacy risk does not simply arise from the fact that data is sensitive, but from the combination of its intrinsic sensitivity and the breadth of the audience that can access it. However, this approach was limited to structured and clearly categorizable profile data, not addressing unstructured textual content generated by users, which was a primary source of unintentional disclosure. Furthermore, it does not consider other categories of features (behavioral and network) that can significantly impact the overall risk assessment.

Complementary research has addressed related issues: privacy risks arising from the establishment of new social relationships, data breaches and privacy violations, mapping of privacy preferences across multiple platforms, and automatic identification of private information in texts using Natural Language Processing. From a behavioral perspective, research has extensively documented the privacy paradox—the discrepancy between stated privacy concerns and actual disclosure behavior—associated with individuals’ limited rationality, structural information asymmetries, and deceptive design patterns strategically implemented by digital platforms.

The studies proposed by Acquisti [16], [25] have provided essential theoretical foundations for understanding how systematic psychological factors, uncertainties, and context manipulations lead to the unwitting disclosure of sensitive information. Formal privacy metrics such as k -anonymity and its variants (l -diversity, t -closeness) have been well-established approaches for protecting data

in published datasets, although their application in real-time privacy risk assessment in social media remains limited due to the difficulty of implementation in dynamic contexts.

2.7 Conclusion

The analysis conducted reveals that privacy, far from being a static concept, today represents a fluid, stratified, and continuously evolving category, confronting increasingly complex challenges. Digital society has enormously expanded the boundaries of what is considered “personal data”, including apparently neutral information, unstructured content, and generic traces that, when combined, become powerful instruments of identification and profiling.

The first critical aspect is the distorted perception that users have of the value and scope of the data they share online. The casualness with which information is provided on social media, facilitated by manipulative interfaces and opaque design logics, generates a continuous flow of data that fuels business models based on profiling, often without full awareness or genuine consent.

The second element concerns the evolution of the concept of privacy itself, which has transitioned from a “right to be let alone” to a structural principle of informational self-determination and a guarantee of fundamental rights. The GDPR and European jurisprudence have made evident that personal data protection is not an end in itself but rather a necessary instrument for preserving dignity, equality, and substantive freedoms.

The third crucial dimension is represented by Personal Identifiable Information (PII), which constitutes the technical-legal foundation of identity construction. The possibility of identifying or re-identifying an individual through direct or indirect data is the prerequisite that enables both economic profiling and political surveillance, accentuating power asymmetries between users and large platforms. Finally, the fourth domain has demonstrated how unstructured and user-generated content – photos, audio, posts – contains a latent informational resource, rich in metadata and implicit meanings. This content represents

the most advanced frontier of privacy risk, because it is difficult to anonymize, easily reusable for inferences, and potentially discriminatory when interpreted by opaque and inequitable algorithms.

In synthesis, this analysis highlights the necessity of a multilevel approach: strengthening of regulations, development of innovative technical solutions, greater algorithmic transparency, platform responsibility, and, no less importantly, enhancement of individual awareness. Only through this approach will it be possible to balance innovation with the protection of fundamental rights, restoring to individuals control over their digital identity and preventing the information society from irreversibly transforming into a surveillance society.

Chapter 3

Related Work

This chapter reviews the main studies and datasets related to the detection of privacy-sensitive content on social media. In particular, it examines the resources most relevant to this thesis, highlighting their characteristics, annotation schemes, research objectives, and limitations as they relate to the thesis’s purpose.

3.1 PII Disclosure Dataset

The dataset introduced by Rosado in [26] presents an extensive study aimed at quantifying the presence of personally identifiable information (PII) on social media. It comprises more than 120,000 Reddit posts, collected and processed within a broader analytical framework aimed at studying information disclosure dynamics in online communities. The dataset is publicly available on Kaggle and includes both post-level and cluster-level information; however the authors did not release the original text.

At the post level, each instance is associated with de-identified unique identifiers for both the post and the user, together with a numerical risk score that quantifies the degree of privacy exposure. This score is computed through

the PII-Codex framework, a custom library based on named entity recognition techniques that identifies sensitive entities within the text. Each detected entity is assigned a risk level on a three-point scale, distinguishing between non-identifiable, indirectly identifiable, and directly identifiable information. The overall risk score of a post is then derived as the average of the token-level scores. In addition to this quantitative measure, the dataset provides information about the categories of disclosed PII, the presence or absence of sensitive content, and temporal metadata.

More specifically, each post is uniquely identified through a de-identified *post_uuid*, while the corresponding author is represented through a de-identified *user_uuid*. These identifiers preserve linkability across observations, allowing posts authored by the same user to be tracked throughout the dataset. Therefore, the data should be considered de-identified rather than fully anonymized. The variable *pii_detected* explicitly indicates whether any personally identifiable information has been identified within the post, whereas *pii_disclosed* stores the list of detected specific PII categories, drawn from a taxonomy comprising 54 distinct types. The dataset also includes a temporal variable, *timestamp*, which records the exact creation time of each post, and a derived variable, *time_elapsed*, which measures the number of seconds elapsed from the origin of the corresponding conversational thread. Contextual information is also provided through the variables *community*, identifying the subreddit in which the post was published, and *cluster_name*, which associates the post with a specific conversational cluster.

In addition to content-related features, the dataset incorporates structural and network-based attributes that enable the analysis of user influence on the platform. In particular, the variable *influence_power* represents the eigenvector centrality of the user within the interaction network, while *user_influencer_tier* categorizes users into hierarchical levels (ranging from non-influencers to mega-influencers) based on their relative centrality and reach. These variables make it possible to investigate how influential users contribute to the dissemination of potentially sensitive information.

To provide a high-level overview, the dataset aggregates data into specific conversational themes aggregating information at the level of conversational clusters. Each cluster is characterized by the total number of posts (*node_count*), the number of posts containing detected PII (*pii_detection_count*), and a set of statistical indicators summarizing the distribution of risk scores within the cluster, including mean, median, minimum and maximum values. Additional aggregate descriptors include the distribution of influencer tiers participating in the conversation (*influencer_tier_frequencies*) and the temporal duration of the interaction (*timestamp_span_sec*), measured as the time span between the first and the last post in the cluster.

This multi-level structure, combining fine-grained post-level annotations with aggregated conversational features, enables the analysis of privacy risk not only as an intrinsic property of textual content, but also as a dynamic phenomenon emerging from user interactions and network structures.

In the literature, this dataset was primarily used in the original study to investigate the relationship between user influence and the propagation of sensitive information. The analysis shows that highly central users tend to amplify the dissemination of PII, highlighting the role of network structure in shaping privacy risks. However, its adoption in subsequent research remains limited, mainly due to its recent publication. More broadly, the dataset aligns with a line of research that equates privacy risk with the presence of identifiable information, focusing on measurable disclosures rather than contextual interpretation.

Despite its richness and scale, the dataset presents several limitations with respect to the objectives of this thesis. The notion of privacy is restricted to explicit PII detection, thus excluding implicit disclosures that arise from linguistic context or require inferential reasoning. Reliance on automated extraction techniques introduces potential inaccuracies and does not capture semantic nuances. Moreover, although user identifiers are available, the dataset is primarily designed to analyze influence dynamics rather than privacy risk. As a consequence, it is more suitable for studying information leakage than for modeling privacy risk as a complex, context-dependent phenomenon.

3.2 SPeDaC Dataset

The SPeDaC dataset, introduced in [27], was developed to address a major limitation in Sensitive Information Detection (SID) research: the absence of a standardized benchmark for identifying semantically complex personal data in natural language. Existing SID research had largely concentrated on narrow forms of Personally Identifiable Information (PII), such as names, phone numbers, or account identifiers, often detected through rule-based systems, regular expressions, or shallow lexical matching approaches. Previous datasets, including the Enron Email Dataset and the Monsanto Papers corpus, were domain-specific, outdated, or not designed for fine-grained privacy classification tasks. In addition, many previous studies focused on structured identifiers rather than contextual privacy semantics. The authors therefore positioned SPeDaC as a benchmark explicitly grounded in contemporary privacy regulation and capable of supporting context-aware classification of personal data in unstructured text.

The conceptual motivation for the dataset emerges from the growing role of privacy-sensitive NLP systems in modern digital environments. Applications such as conversational agents, customer support systems, social media platforms, and human-robot interaction increasingly process sensitive personal information, creating significant risks of privacy leakage and regulatory non-compliance. The paper situates its contribution within the broader literature on privacy-preserving data mining (PPDM), data loss prevention (DLP), and sensitive information detection. In particular, the authors highlight that while many PPDM approaches focus on sanitization, anonymization, perturbation, or cryptographic protection after sensitive information has already been identified, a fundamental prerequisite remains the reliable automatic detection of sensitive content itself. Consequently, SPeDaC was designed as a resource for evaluating computational systems capable of detecting sensitive personal data before disclosure or leakage occurs.

The dataset is theoretically grounded in the Data Privacy Vocabulary (DPV) created by the World Wide Web Consortium (W3C), a machine-readable ontology aligned with the definitions of personal data established by the Gen-

eral Data Protection Regulation (GDPR). The GDPR defines personal data as any information relating directly or indirectly to an identifiable individual, including identifiers related to physical, physiological, genetic, mental, economic, cultural, or social identity. Based on this regulatory definition, the authors adopted the DPV-PD extension, which contains 206 Personal Data Categories (PDCs). However, rather than using the ontology directly, the researchers conducted a conceptual filtering process to identify categories that are realistically expressible through natural language.

This process led to a hierarchical classification that distinguishes the categories based on their meaning and computational feasibility. The authors identified five broad macro-categories of personal data used in the dataset: Special Category Data, Financial and Tracking, Social, Internal, and External. These categories derive from the DPV hierarchy and group together semantically related forms of personal information. Special Category Data includes highly sensitive attributes regulated under GDPR Article 9, such as health conditions, ethnicity, political opinions, religion, disability, and sexual orientation. The financial and Tracking categories include monetary information, ownership, accounts, contact information, and location tracking. Social categories concern family, relationships, professional networks, and public life. Internal categories refer to psychological or preference-related information, such as beliefs, intentions, or favorite activities, whereas External categories describe observable characteristics such as physical appearance or behavioral traits.

The authors further distinguished between categories identifiable through contextual textual analysis and categories more easily detectable through fixed patterns. Their primary focus was on semantically complex categories that required contextual interpretation. For example, age can be expressed in multiple indirect forms (“I’m seventeen,” “I was born in 2005,” “I was only five in 2010”), making purely lexical identification insufficient. In contrast, categories such as credit card numbers or tax identifiers were considered less challenging because they can often be recognized through regular expressions. This distinction reflects one of the central arguments of the paper: sensitive information detection should not be reduced to entity extraction but instead should be approached as

a context-dependent semantic classification problem.

The texts used to construct SPeDaC were extracted from the English TenTen2020 corpus, a large-scale web corpus containing approximately 36 billion words collected between 2019 and 2021 from multiple English-speaking regions and web domains. The corpus includes heterogeneous sources such as news articles, blogs, online discussions, legal texts, scientific pages, and Wikipedia content. This diversity was intended to maximize lexical and contextual variability while maintaining contemporary language usage. Importantly, the dataset was constructed exclusively at the sentence level. No user metadata, conversational history, profile information, or interaction context was retained. This design decision enables controlled experiments on isolated textual units but simultaneously abstracts away from the dynamics of real online interaction.

The dataset consists of three distinct modules, each addressing a different classification challenge. SPeDaC 1 is a binary classification dataset containing 10,675 sentences labeled as sensitive or non-sensitive. The objective is to determine whether a sentence contains personal data. SPeDaC 2 introduces multiclass classification over five macro-categories and contains 5,133 sensitive sentences balanced across categories. SPeDaC 3 constitutes the most granular version of the dataset and includes 5,562 sentences annotated with 61 fine-grained personal data categories. These include categories such as marital status, criminal history, income bracket, professional evaluation, ethnicity, religion, political affiliation, location, family structure, health status, favorite music, and many others.

One of the most innovative aspects of SPeDaC is its explicitly context-aware annotation methodology. The authors argue that sensitivity cannot be determined solely through keywords because the same linguistic expression may or may not be sensitive depending on contextual usage. Consequently, the dataset intentionally includes adversarial examples in which sensitive and non-sensitive sentences share nearly identical lexical structures. For example, a sentence mentioning a profession or medical condition may be genuinely disclosive in one context but hypothetical, fictional, quoted, or ironic in another. The dataset

therefore forces models to perform semantic interpretation rather than memorizing lexical correlations.

To achieve this objective, the annotation process relied on two forms of linguistic constraints. General constraints focused on the relationship between personal information and identifiable subjects, especially first-person references suggesting self-disclosure. Specific constraints targeted linguistic constructions associated with particular PDCs, such as expressions indicating age or employment. Non-sensitive adversarial examples were then generated using contextual modifiers such as quotations, assumptions, jokes, hypothetical statements, or indirect speech. This methodology makes SPeDaC substantially more difficult than traditional PII datasets because contextual ambiguity becomes central to classification performance.

The annotation process itself was conducted manually using the INCEPTION annotation platform and the WebAnno TSV v3.3 format. Multiple annotators with linguistic expertise participated in validation tasks using the DPV taxonomy as a reference framework. The agreement between annotators was evaluated using Krippendorff’s alpha coefficient, chosen because it is robust to multiple annotators and missing labels. The reported agreement scores were 0.73 for SPeDaC 1, 0.82 for SPeDaC 2, and 0.87 for SPeDaC 3, indicating substantial reliability despite the semantic complexity of the task. Interestingly, disagreement was higher in binary sensitivity classification because annotators often differed on whether ambiguous statements should count as personal disclosures.

The paper also presents an extensive benchmark evaluation that compares classical machine learning models and transformer-based architectures. Traditional models included k-Nearest Neighbors (k-NN), Support Vector Machines (SVM), and Logistic Regression (LR), while deep learning approaches included RoBERTa, DeBERTa, and LaBSE. Classical models used bag-of-words representations and handcrafted preprocessing pipelines involving tokenization, lemmatization, lowercase normalization, and stop-word removal. Transformer architectures instead exploited contextual embeddings and attention mechanisms

capable of modeling semantic dependencies within sentences.

The experimental results demonstrate the superiority of contextual transformer models for privacy classification tasks. RoBERTa achieved the highest accuracy on SPeDaC 1, exceeding 90% in binary sensitivity detection, while DeBERTa obtained the best performance on SPeDaC 2. These results confirm that contextualized attention mechanisms are particularly effective when semantic ambiguity plays a central role. However, all models experienced substantial performance degradation on SPeDaC 3, where the task required discrimination among 61 fine-grained categories with limited training examples per class. Even advanced transformer architectures struggled with semantically similar categories belonging to the same macro-domain. The paper therefore identifies fine-grained privacy classification as an open research challenge.

The influence of SPeDaC rapidly extended beyond its original publication, and the dataset became an important benchmark for privacy-aware NLP, confidentiality classification, multilingual SID, and AI governance research. Several subsequent studies adopted or referenced SPeDaC in order to evaluate privacy classification systems and investigate semantic sensitivity in language models. Chen [28] discuss SPeDaC within a broader review of information sensitivity in digital platforms and AI-driven systems, identifying it as one of the most relevant resources for studying semantically complex privacy disclosures using NLP techniques. Sulavko [29] similarly use SPeDaC as a benchmark for confidentiality classification with neural network ensembles and emphasize the strong performance of transformer architectures such as RoBERTa in the SPeDaC 1 binary classification task.

The dataset also influenced the multilingual privacy research. Li [30] developed the SeDaC corpus for Chinese sensitive information detection by explicitly adapting the hierarchical annotation methodology and DPV-PD taxonomy introduced by SPeDaC and their work reproduces the distinction between genuinely sensitive disclosures and semantically ambiguous or adversarial examples, thus extending the context-aware philosophy of SPeDaC to multilingual settings. In cybersecurity and cloud security research, Sarumathy [31] describe SPeDaC

as a structured benchmark for evaluating AI-based intrusion prevention systems and privacy-aware cloud infrastructures, highlighting the usefulness of its multi-level taxonomy for increasingly granular classification tasks. Finally, De Renzis [32] reference SPeDaC while studying the use of large language models to automatically generate sensitive information detectors, stressing the importance of manually curated privacy datasets for robust evaluation.

The authors also performed error analyzes and embedding visualizations using t-SNE representations of transformer states. Their analyzes revealed that many classification errors emerged from sentences containing multiple overlapping personal data categories or from semantically ambiguous contexts. For example, a sentence simultaneously mentioning marital status and geographic location may require multi-label annotation, whereas the dataset currently assumes a single dominant label. The authors therefore propose future extensions involving token-level BIO annotation schemes and multi-label classification frameworks.

From a broader perspective, SPeDaC contributes significantly to privacy-aware natural language processing because it operationalizes privacy according to formal legal and ontological frameworks rather than superficial identifiers alone. Unlike many prior SID resources, the dataset explicitly models semantic contextuality and adversarial ambiguity, making it particularly valuable for evaluating robust privacy detection systems. Furthermore, by aligning annotations with GDPR and DPV definitions, the resource establishes a bridge between computational NLP research and legal privacy regulation.

Nevertheless, the dataset also exhibits important limitations. First, because all instances are isolated sentences, the corpus lacks conversational continuity, temporal evolution, and interactional dynamics. Privacy in real social media environments often emerges cumulatively through sequences of disclosures rather than isolated utterances. Second, the absence of user metadata prevents the modeling of relational privacy, identity linkage, or audience-dependent disclosure risks. Third, the dataset focuses on semantic categorization rather than explicit privacy risk assessment: it identifies which type of personal data is present but

does not estimate the severity, exploitability, or contextual risk associated with disclosure. Finally, while the dataset captures linguistic ambiguity, it remains relatively artificial compared to naturally evolving social interactions.

Consequently, SPeDaC is best understood as a highly controlled benchmark for taxonomy-driven sensitive information detection and privacy classification rather than a realistic simulation of privacy risk in online social systems. Its principal contribution lies in reframing privacy detection as a context-sensitive semantic classification problem grounded in formal regulatory definitions, thus providing an essential benchmark for future research in privacy-aware NLP, sensitive information detection, and data protection technologies.

3.3 Privacy Leaks in Tweets Dataset

The dataset proposed by Cappellari in [33] was developed in the context of detecting privacy leaks in social media posts, with a specific focus on Twitter. It represents one of the early attempts to construct an annotated dataset grounded in a perception-based definition of privacy, aiming to support supervised machine learning approaches for privacy classification.

The dataset consists of approximately 6,000 tweets collected from the Twitter streaming API over multiple periods between 2017 and 2018. To ensure data quality and a broad scope, a filtering process was applied: only English-language tweets were retained, and a minimum length threshold of at least 10 words (excluding stop words) was enforced. In addition, hashtags, URLs, and retweets were removed. These constraints were introduced to obtain self-contained textual messages that do not rely on external context or platform-specific metadata, thus making the classification task more robust and domain-independent.

The annotation process was carried out using Amazon Mechanical Turk (AMT), with each tweet evaluated by five independent annotators. The labeling task was designed to capture a “societal” notion of privacy, reflecting how users perceive sensitive information in practice rather than relying on predefined legal or technical categories. Annotators were asked to classify each tweet

into one of three categories: private, somewhat private, or not private. The “somewhat private” category was introduced to capture ambiguous cases where the sensitivity of the information is uncertain or subjective.

For the final dataset used in modeling, the annotations were aggregated using a majority voting scheme. Specifically, if at least 60% of the annotators labeled a tweet as private (including the somewhat private category, which was merged into private), the message was assigned to the private class; otherwise, it was labeled as not private. This transformation results in a binary classification problem better suited for machine learning models.

The resulting dataset includes anonymized tweet text and confidence-related information derived from inter-annotator agreement. However, it does not provide structured metadata regarding the specific types of sensitive information disclosed, nor does it include detailed conversational or contextual information. This limits its use for fine-grained privacy risk analysis.

In the literature, this dataset has been used to train and evaluate supervised models for privacy leak detection and to study patterns of sensitive information disclosure in microblogging platforms. Its main contribution lies in the adoption of a perception-driven definition of privacy, which captures collective human judgment rather than relying on static or rule-based definitions.

However, several limitations should be noted. First, the subjective nature of the annotations introduces variability and potential inconsistency across examples, which may affect model stability. Second, the relatively small size of the dataset limits its suitability for modern deep learning approaches that require large-scale training data. Third, the absence of a structured taxonomy of sensitive information types restricts the ability to perform detailed categorization of privacy leaks. Finally, limited accessibility of the dataset may hinder reproducibility and comparative evaluation across studies.

Overall, while the dataset provides valuable insights into perceived privacy in social media, it is primarily suitable for exploratory analysis and baseline machine learning tasks, rather than comprehensive or high-resolution privacy

risk modeling.

The dataset proposed by Cappellari rapidly became one of the first benchmarks for privacy-sensitive text classification on social media platforms, particularly Twitter. Several subsequent studies adopted the dataset or referenced its annotation methodology in order to evaluate automatic privacy detection systems, compare machine learning architectures, and investigate contextual privacy leakage in online communication.

Guarino [34] discuss the Cappellari dataset within the broader context of privacy-awareness systems for social media platforms. The authors describe the original binary classification framework proposed by Cappellari, where tweets were manually annotated as private or non-private and classified using supervised learning approaches such as Support Vector Machines (SVM), Random Forests (RF), and Naive Bayes classifiers. Their analysis highlights that SVM achieved the strongest performance among the evaluated models, obtaining approximately 70% accuracy using TF-IDF representations.

The same work additionally emphasizes the methodological contribution of the larger 6,000-tweet corpus later constructed through Amazon Mechanical Turk (AMT) annotation campaigns. Guarino underline how the dataset enabled the comparison of multiple machine learning techniques, including Generalized Linear Models, Decision Trees, Boosting, Bagging, and Supervised Latent Dirichlet Allocation (SLDA). Furthermore, the authors contrast the topic modeling strategy employed by Cappellari, based on Latent Dirichlet Allocation (LDA), with their own browser-based privacy awareness system designed for real-time user assistance.

Additional evidence of the dataset’s influence appears in recent deep learning research on privacy detection. Niu [35] identify Cappellari’s dataset as one of the foundational works in automatic privacy leak detection on social media text. Their work situates the dataset within a broader family of privacy classification approaches that include both traditional machine learning and transformer-based architectures. In particular, the authors note that previous studies based on the Cappellari corpus mainly focused on original social media posts rather

than conversational replies or contextual interactions.

Niu further argue that existing privacy classifiers, including those evaluated by Cappellari, remain primarily optimized for isolated tweet classification tasks. Consequently, their own work extends the problem toward privacy violation detection in social media comments and conversational interactions, where contextual dependencies become substantially more important. This observation highlights one of the principal limitations of early privacy detection benchmarks: the absence of conversational context and longitudinal interaction dynamics.

Overall, the continued citation of the Cappellari dataset across privacy-aware NLP, social media analysis, and deep learning research confirms its historical importance as one of the earliest supervised benchmarks for privacy-sensitive tweet classification. The literature increasingly recognizes the dataset not only as a benchmark for binary privacy detection, but also as a foundational resource that motivated later developments in contextual privacy inference and conversational privacy analysis.

3.4 Privacy-Sensitive Content Dataset

The dataset introduced by Bioglio and Pensa [36] represents one of the first attempts to explicitly model privacy-sensitive content in user-generated text without relying on proxy assumptions such as anonymity or visibility settings. Unlike prior approaches that equate privacy with posting behavior (e.g., anonymous vs. identified users), this work proposes a direct and content-based definition of privacy sensitivity, grounded in the notion of information disclosure.

Formally, a piece of content is defined as privacy-sensitive if it discloses, either explicitly or implicitly, personal information about the author or other identifiable individuals. This definition extends beyond traditional notions of self-disclosure, as it also includes information about third parties and implicitly conveyed personal attributes. For example, references to habits, health conditions, movements, or social relationships are considered sensitive even in the absence of explicit identifiers. This conceptualization directly addresses limita-

tions highlighted in prior literature, where sensitivity was often approximated through indirect signals such as anonymity [37], [38] or restricted to specific domains such as health support forums [39], [40].

The dataset is built upon a corpus of 9,917 Facebook posts collected within the myPersonality project, a widely used resource in computational social science. All posts were manually sanitized to remove explicit identifiers (e.g., personal names), while preserving semantic content. Compared to datasets derived from platforms such as Twitter or Reddit, Facebook posts are considered more suitable for this task, as they are associated with more realistic self-representation behaviors and a higher likelihood of genuine personal disclosure.

The annotation process involved twelve expert annotators, organized into groups of three, with each post receiving exactly three independent annotations. Annotators were provided with detailed guidelines defining four possible labels: *sensitive*, *non-sensitive*, *unknown*, and *unintelligible*. A post is labeled as sensitive if it clearly contains information that may violate a person’s privacy, including (but not limited to) health status, personal relationships, daily activities, geolocational cues, or political and religious views. Importantly, the guidelines explicitly account for contextual and implicit disclosures, reflecting the observation that privacy risk is not always tied to explicit lexical cues.

The annotation statistics highlight the intrinsic difficulty of the task. Agreement measured using Fleiss’ κ ranges from fair to moderate, confirming that privacy sensitivity is inherently subjective and context-dependent. To address this variability, two filtered datasets were derived: SENS2, containing posts with at least two annotators in agreement, and SENS3, including only unanimously labeled instances. This design enables controlled experimentation under different levels of annotation reliability.

In the literature, this dataset has primarily been used in the original study to benchmark machine learning models for sensitivity classification. The authors demonstrate that sequential deep neural architectures outperform traditional bag-of-words approaches, highlighting the importance of capturing contextual

and semantic information. Moreover, the study shows that models trained under the common “anonymity assumption” perform poorly when applied to this dataset, providing empirical evidence against the validity of anonymity as a proxy for privacy sensitivity.

More broadly, this work positions itself within a line of research that critiques existing approaches to privacy modeling. Previous studies have relied on lexicon-based methods [41], anonymity signals [37], or platform-specific behaviors, while others have focused on text sanitization through named entity recognition and knowledge bases. In contrast, the Privacy-Sensitive Content dataset introduces a more general and flexible framework that captures both explicit and implicit disclosure, making it particularly relevant for studying realistic privacy risks in open-domain social media.

Despite these strengths, several limitations must be acknowledged. The dataset remains relatively small compared to modern large-scale corpora, which constrains the applicability of data-intensive deep learning models. Additionally, the absence of user-level metadata and interaction context prevents the analysis of cumulative privacy exposure and longitudinal behaviors. Finally, the binary nature of the sensitivity label (after filtering) limits the possibility of modeling different levels of privacy risk.

Nevertheless, the dataset constitutes a foundational resource for the study of privacy as a semantic and contextual phenomenon. Its explicit definition of sensitivity, combined with high-quality manual annotation, makes it particularly suitable for tasks that require going beyond surface-level detection of personally identifiable information and toward a more nuanced understanding of privacy risk.

3.5 PersonalReddit Dataset

To address how Large Language Models (LLMs) exploit indirect personal data, [42] introduced PersonalReddit, a dataset specifically designed to evaluate LLM-driven privacy leaks through text inference. This objective reflects a broader

shift in privacy research: while traditional studies mainly focused on explicit personally identifiable information (PII), memorization of training data, or data leakage, the PersonalReddit dataset addresses a more subtle and increasingly relevant risk, namely implicit privacy leakage through inference.

The motivation behind the dataset stems from the observation that recent LLMs possess strong contextual reasoning and world knowledge capabilities. Such models are often able to infer information that users never intended to reveal explicitly. For instance, a comment mentioning a specific transportation practice, regional slang, educational structure, or cultural custom may indirectly reveal the user’s geographical origin, age group, or social background. The authors argue that these inference capabilities significantly lower the cost of large-scale automated profiling, enabling malicious actors to infer sensitive information without requiring explicit identifiers or sophisticated hand-crafted profiling systems.

The dataset consists of 520 Reddit user profiles associated with a total of 5,814 comments collected between 2012 and early 2016. The comments were extracted from the PushShift Reddit archive and filtered according to several criteria. Only comments written by non-deleted users and containing sufficient textual content were retained. Furthermore, the authors restricted the collection to 381 subreddits considered likely to contain personal disclosures or contextual references related to users’ lives, occupations, demographics, or social identities. The selection of Reddit as the source platform is particularly relevant because Reddit combines large-scale public textual interaction with a semi-anonymous environment in which users often share personal experiences more openly than on traditional social networks.

Unlike many previous author profiling datasets, which typically focus on a small number of demographic attributes such as age or gender, PersonalReddit adopts a much broader notion of personal information inspired by census research and legal definitions of personal data. Each profile is manually annotated with up to eight demographic and socio-economic attributes:

- age;

- gender;
- education level;
- occupation;
- marital or relationship status;
- current location;
- place of birth;
- income category.

This broader attribute space reflects the authors’ intention to investigate privacy risks beyond classical PII extraction. In particular, several of these attributes fall under legal definitions of sensitive personal data because they can contribute to identifying individuals or constructing highly detailed personal profiles when combined with other publicly available information.

A particularly important aspect of the dataset is the inclusion of additional meta-annotations that characterize the inference process itself. For every labeled attribute, annotators assigned:

- a **certainty score**, indicating how confident they were that the inferred label was correct;
- a **hardness score**, ranging from 1 to 5, measuring how difficult it was to infer the attribute from the textual evidence.

These annotations are crucial because they provide insight into the complexity of privacy inference tasks. Hardness level 1 corresponds to nearly explicit disclosures, where the attribute can be directly extracted from the text. In contrast, hardness level 5 corresponds to highly indirect inferences requiring substantial contextual reasoning, external cultural knowledge, or interpretation of subtle linguistic patterns. For example, a user mentioning “hook turns” in traffic discussions may indirectly reveal residence in Melbourne, since that traffic maneuver is strongly associated with that city. Similarly, references to

educational systems, transportation zones, regional food, sports teams, or local expressions may reveal geographical or socio-economic information without any direct disclosure.

The inclusion of hardness annotations allows the dataset to distinguish between different forms of privacy leakage. Traditional privacy research generally focuses on explicit leakage, where sensitive information is directly stated. PersonalReddit instead demonstrates that modern LLMs can exploit latent contextual signals embedded in ordinary language use. This distinction is fundamental because inferential privacy risks are much more difficult to mitigate using conventional anonymization approaches.

The annotation process was performed entirely by the authors of the study rather than outsourced to external annotators. This decision was motivated by ethical concerns related to the handling of sensitive personal information. Annotators manually reviewed Reddit profiles and extracted labels according to detailed annotation guidelines. For more difficult cases (hardness levels 4 and 5), annotators were allowed to use external search engines to contextualize references found in the comments, although the use of LLMs during annotation was explicitly prohibited. The complete annotation process required approximately 112 hours of manual work.

To improve label quality and consistency, part of the dataset was cross-labeled by multiple annotators. The reported agreement exceeded 90% for labels associated with medium-to-high certainty levels, indicating relatively strong consistency despite the inherent ambiguity of some inference tasks. The authors also restricted the main evaluation to labels with certainty scores of at least 3 in order to reduce noise and avoid evaluating models on highly uncertain annotations.

Another relevant contribution of the dataset is its explicit consideration of memorization risks. Since privacy concerns in LLMs are often associated with verbatim memorization of training data, the authors performed a decontamination study to verify whether the comments contained in PersonalReddit had been memorized by the evaluated models. The study showed that memoriza-

tion effects were negligible, apart from a few common URLs, jokes, or widely repeated online expressions. This result is important because it supports the conclusion that the observed privacy violations primarily arise from inferential reasoning rather than from memorized training examples.

The dataset was used to evaluate the capabilities of several state-of-the-art LLMs, including GPT-4, GPT-3.5, Claude, PaLM-2, and multiple Llama-2 variants. The experiments revealed remarkably high inference performance across all evaluated attributes. GPT-4 achieved approximately 85.5% top-1 accuracy and more than 95% top-3 accuracy on labels with sufficient certainty. The results were particularly striking because the models often inferred attributes correctly even when humans required additional contextual information or external searches. For example, GPT-4 demonstrated strong performance in location inference tasks, correctly identifying geographical regions from subtle contextual references or region-specific linguistic cues.

The experiments also revealed a strong correlation between model scale and inference capability. Larger models consistently outperformed smaller ones across nearly all attributes and hardness levels. This trend suggests that advances in language understanding directly increase the ability of models to infer private information, raising important concerns regarding the future scalability of automated profiling systems.

A major contribution of the study concerns the evaluation of privacy mitigation techniques. The authors tested commercial anonymization systems, particularly Azure Language Services, to determine whether removing explicit identifiers would prevent attribute inference. The anonymization process removed recognized entities such as names, locations, organizations, dates, numbers, and other forms of structured personal information. However, the results demonstrated that anonymization was only partially effective. While performance decreased for easy examples containing explicit references, models still achieved relatively high accuracy on more difficult samples. In particular, GPT-4 retained strong inference capabilities even after anonymization, especially for attributes such as location, age, and place of birth.

This finding highlights one of the most important implications of the PersonalReddit dataset: privacy risks associated with LLMs extend beyond explicit leakage of PII. Modern models are capable of leveraging contextual clues, discourse patterns, and implicit semantic associations that remain present even after traditional anonymization techniques have removed direct identifiers. Consequently, current anonymization methods may be insufficient in environments where advanced LLMs are used for profiling or surveillance purposes.

The dataset also allowed the authors to explore an additional threat model involved adversarial conversational agents. In this setting, malicious chatbots subtly steer conversations in order to elicit responses containing enough contextual information to infer private attributes. Although the experiments were simulated for ethical reasons, they demonstrated that current LLMs can already conduct highly effective privacy-invasive interactions while appearing to engage in ordinary conversation. This aspect further expands the relevance of PersonalReddit beyond passive text analysis toward interactive AI systems.

Despite its importance, the dataset has several limitations. The most significant limitation concerns reproducibility. Due to ethical and privacy considerations, the original dataset is not publicly available. Instead, the authors provide only synthetic examples designed to mimic the structure and characteristics of the real data. Although, this approach protects user privacy, it restricts independent verification and benchmarking by the research community.

Another limitation is that the dataset primarily measures attribute inference accuracy rather than privacy harm directly. In other words, it evaluates whether models can infer specific attributes, but it does not quantify the sensitivity, contextual impact, or potential societal consequences of those inferences. Therefore, the dataset provides an indirect perspective on privacy risk, focusing mainly on inferential capability rather than on broader notions of privacy exposure or vulnerability to users.

Additionally, the dataset is limited to Reddit users and English-language online discussions from a specific temporal period. Since linguistic practices, online cultures, and platform behaviors vary substantially between communi-

ties and over time, the generalizability of the findings to other platforms or languages remains uncertain. The selected subreddits were also intentionally biased toward communities likely to contain personal disclosures, which may overestimate inference performance compared to more neutral online settings.

Nevertheless, despite these limitations, PersonalReddit represents one of the most influential and comprehensive datasets currently available to study inference-based privacy threats in large language models. Its main contribution lies in demonstrating that modern AI systems can reconstruct highly sensitive personal profiles from ordinary textual interactions, even in the absence of explicit identifiers. The dataset therefore shifts the discussion of privacy in AI from simple data leakage toward a broader understanding of privacy as a problem of inference, contextual reasoning, and automated profiling.

The work proposed by Staab in [42] rapidly became an important reference in the literature on inferential privacy risks and attribute inference attacks involving large language models. Several recent surveys on LLM security and privacy refer to the work as one of the first systematic analyzes of attribute inference risks in generative language models. For example, LLM privacy and security surveys describe attribute inference attacks as a major emerging threat in which attackers reconstruct sensitive user information through contextual reasoning and model output [43], [44]. These surveys explicitly cite experiments conducted by Staab on Reddit profiles, where language models successfully inferred attributes such as location, income, sex, and political orientation.

The impact of the work also extends to research on AI-driven persuasion and personalization. Salvi [45] reference the study while discussing how modern LLMs can infer psychological and demographic traits from digital traces, allowing increasingly sophisticated forms of personalized persuasion and micro-targeting.

Additional works connects these findings with broader concerns about memorization and learning of neural representation. Behrouz [46] cite the work when discussing how memorization mechanisms in neural architectures may introduce privacy vulnerabilities and unintended information leakage.

In general, the paper contributed to shifting privacy research from explicit information leakage to inferential privacy risks, where sensitive attributes emerge through contextual aggregation, semantic reasoning, and behavioral analysis rather than direct disclosure alone.

3.6 SynthPAI Dataset

The SynthPAI [47] dataset represents a recent benchmark specifically designed to study privacy inference risks in online conversational environments through the use of synthetic yet behaviorally realistic user profiles. Unlike traditional datasets collected from real social media platforms, SynthPAI is generated through a fully controlled pipeline in which each artificial user is assigned a pre-defined set of demographic and personal attributes before any textual content is produced. This design enables the creation of large-scale conversational data while maintaining exact ground-truth annotations for all sensitive attributes.

Each synthetic profile is characterized by multiple personal dimensions, including age, place of birth, current location, education level, income level, occupation, relationship status, and sex. Age is represented as a continuous variable ranging from 18 to 99 years, while geographic information is modeled through city-country tuples that describe both birthplace and current residence. Education and occupation are generated as free-text descriptions to mimic realistic user self-expression, although education is later mapped into standardized categories such as high school, bachelor’s degree, master’s degree, and PhD for evaluation purposes. Income is initially generated as a continuous salary estimate in local currency and subsequently normalized into categorical levels (low, medium, high, very high) according to regional socioeconomic standards.

A central characteristic of SynthPAI is the use of persona-driven conversational generation. Instead of explicitly embedding sensitive information into comments, the dataset relies on contextual leakage, implicit linguistic patterns, and behavioral consistency between conversations. Synthetic agents participate in Reddit-style discussion threads generated using GPT-4-based prompt-

ing strategies. Each agent interacts according to its assigned persona, producing comments that indirectly reveal demographic and personal information through writing style, topical preferences, cultural references, and conversational behavior.

The thread generation procedure is designed to reproduce realistic online interaction dynamics. Threads are generated through multiple conversation rounds, with users probabilistically deciding whether to reply, continue an existing subdiscussion, or disengage from the conversation. The generation process explicitly models user engagement patterns by assigning higher probabilities to replies within already visited subthreads while reducing engagement likelihood as discussion depth increases. This behavior is formalized through a scoring function that accounts for thread visibility, prior participation, and conversational depth.

Several hyperparameters are used to control generation quality and realism. During dataset construction, threads were generated with a maximum depth of five levels, up to three responses per comment, and approximately forty participating profiles per discussion. Generated comments vary substantially in length, with a high probability of short informal responses but occasional long argumentative contributions. In addition, a disagreement coefficient was introduced to encourage contrasting opinions and simulate authentic online discourse.

The resulting dataset exhibits realistic structural properties similar to real-world Reddit conversations. Comment length distributions show a bimodal pattern, with peaks corresponding to short replies and longer explanatory comments. Most threads contain fewer than forty participants and approximately eighty to one hundred comments. Discussion chains remain relatively shallow, typically between three and five levels deep, reflecting common patterns of social media interaction .

SynthPAI also emphasizes demographic and topical diversity. The dataset contains more than 160 unique geographic location tuples and more than 220 distinct occupations spanning academia, healthcare, engineering, education, creative industries, public administration, and manual labor. The countries rep-

resented include the United States, Australia, China, Japan, Germany, France, Canada, Brazil, Italy, and the United Kingdom. Relationship status and income categories are distributed relatively evenly, while educational backgrounds consist primarily of bachelor’s and master’s degrees.

To increase topical realism, discussion threads are softly conditioned on subreddit-style communities. GPT-4 is used to associate each generated discussion with thematic categories inspired by real Reddit forums such as r/relationships, r/urbanplanning, r/personalfinance, r/careerguidance, and r/askacademia. This approach ensures that sensitive attributes emerge naturally within semantically coherent conversational contexts rather than through artificial prompts.

An important contribution of SynthPAI is the introduction of hardness labels to represent the difficulty in inferring sensitive attributes from user comments. Hardness levels range from direct and easily inferable signals to subtle contextual cues that require world knowledge and complex reasoning. Experimental evaluations show that model accuracy generally decreases as hardness increases, although larger language models remain capable of inferring many attributes even under highly indirect conditions.

The dataset has been evaluated using a wide range of state-of-the-art language models, including GPT-4, Claude-3, Gemini, Llama-2, Llama-3, Qwen, Mixtral, Mistral, Gemma and Yi models. In the evaluation framework, all comments written by a synthetic user are aggregated and provided to the model, which must infer the underlying personal attributes through step-by-step reasoning and ranked predictions. The results demonstrate that modern large language models can recover substantial personal information even when explicit identifiers are removed.

To mitigate direct privacy leakage, SynthPAI applies automated anonymization procedures based on Azure Language Services. Personally identifiable entities such as names, locations, organizations, phone numbers, email addresses, URLs, IP addresses, and numerical identifiers are removed or replaced with placeholder symbols. Nevertheless, experimental findings reveal that anonymiza-

tion alone is insufficient to prevent attribute inference, as language models can exploit implicit behavioral and stylistic signals embedded within conversational text.

From a literature perspective, SynthPAI represents an important methodological shift in privacy risk assessment research. Earlier datasets such as Privacy Leaks in Tweets or the Privacy-Sensitive Content Dataset mainly framed privacy as a classification problem operating at the level of isolated textual instances. In contrast, SynthPAI was explicitly designed to evaluate privacy leakage emerging from multi-turn conversational behavior and cross-comment contextual aggregation.

The dataset has primarily been used to benchmark the inferential capabilities of modern large language models under controlled conditions. In particular, associated studies investigate whether LLMs can reconstruct demographic and socio-economic attributes from indirect conversational evidence rather than explicit disclosure. This makes SynthPAI particularly relevant within the recent literature on inference-based privacy attacks, where the central threat model is no longer the extraction of identifiers, but the automated profiling capabilities of foundation models.

Compared to PersonalReddit, which relies on real user data and manually inferred annotations, SynthPAI introduces a fully controllable experimental framework. This enables systematic manipulation of demographic distributions, conversational structures, leakage hardness, and anonymization strategies. As a consequence, the dataset has been mainly adopted for the comparative evaluation of large language models, privacy attacks, and anonymization robustness under reproducible conditions.

The literature employing SynthPAI consistently highlights three major findings. First, modern LLMs remain capable of inferring sensitive attributes even after explicit identifiers are removed. Second, conversational consistency and stylistic coherence provide strong latent signals that can be exploited for profiling. Third, privacy risks emerge cumulatively between interactions rather than isolated disclosures alone. These observations reinforce a growing consensus in

recent literature: privacy leakage in LLM-based systems increasingly depends on contextual semantic inference rather than explicit PII exposure.

For the purposes of this thesis, SynthPAI is particularly relevant because it directly models user-level privacy inference in conversational environments. Unlike datasets focused exclusively on sensitive text classification, SynthPAI enables the evaluation of how multiple comments jointly contribute to reconstructing personal profiles. This characteristic aligns closely with the broader objective of studying privacy risk as an emergent property of user behavior rather than as a binary property of individual textual instances.

In general, SynthPAI provides a controlled yet realistic framework for studying privacy risks in conversational AI systems. Its main strength lies in the combination of exact ground-truth annotations, diverse demographic profiles, realistic discussion dynamics, and controlled attribute leakage. At the same time, the synthetic nature of the dataset introduces potential limitations related to realism, cultural bias, and generalizability to real-world social media platforms. Consequently, while SynthPAI constitutes a powerful benchmark for privacy research and adversarial inference evaluation, it should ideally be complemented with real human-generated datasets when assessing practical deployment risks.

The SynthPAI dataset proposed by Yukhymenko in [47] attracted attention in recent research on privacy-preserving language models, anonymization, and personal attribute inference. The dataset was specifically designed to support controlled experimentation on inferential privacy risks by generating synthetic user profiles associated with realistic textual traces.

Subsequent research used SynthPAI as a benchmark for evaluating anonymization and privacy protection strategies in large language models. For example, Kim [48] used the dataset to evaluate adversarial anonymization techniques and self-refined language model anonymizers. Their work exploited the synthetic profiles and associated comments provided by SynthPAI to simulate attacks in which LLMs attempt to infer sensitive user attributes such as age, gender, occupation, education level, and location from textual content. The authors particularly emphasized the difficulty of anonymizing contextual personal infor-

mation compared to explicit identifiers.

The dataset has also been discussed within broader debates concerning synthetic data generation and privacy evaluation methodologies. Venkit [49] referenced SynthPAI while analyzing the limitations and advantages of synthetic benchmarks for attribute inference and deanonymization tasks. Their discussion highlights how synthetic datasets provide controlled experimental settings for evaluating privacy risks, although they may not perfectly reproduce the complexity of real-world online behavior.

More generally, the emergence of datasets such as SynthPAI reflects a growing research trend toward studying inferential privacy threats in conversational AI systems. Unlike traditional privacy datasets focused on explicit personally identifiable information, SynthPAI enables the analysis of latent attribute inference, contextual profiling, and behavioral privacy leakage through realistic language generation scenarios.

Chapter 4

Methodology

This chapter presents the methodological framework adopted for the privacy-risk classification task. It first describes the textual representations used to encode the documents, then introduces the classification models employed in the experiments, and finally outlines the experimental protocol, hyperparameter optimization strategy, and evaluation metrics used to assess model performance.

4.1 Text Representation

To investigate privacy-risk classification from different modeling perspectives, both traditional lexical representations and modern semantic representations are considered. This choice enables a comprehensive comparison between sparse feature-based approaches and dense contextualized representations.

4.1.1 Frequency-Based Representations

The first group of representations relies on lexical information extracted directly from the corpus.

Bag of Words. Bag of Words (BoW) represents each document as a vector of token occurrence counts over the vocabulary observed in the training set. Each dimension corresponds to a unique term, while the associated value indicates the number of times that term appears in the document. Although BoW ignores word order, syntax, and semantic relations, it remains one of the most widely adopted baselines in text classification due to its simplicity, interpretability, and effectiveness.

The resulting representation is typically sparse and high-dimensional, as each document is encoded with respect to the entire vocabulary. Nevertheless, the explicit modeling of lexical evidence often provides strong discriminative power, especially in supervised classification tasks where specific keywords or lexical patterns are strongly associated with the target labels.

TF-IDF. Term Frequency-Inverse Document Frequency (TF-IDF) extends the BoW representation by weighting each term according to its relevance within a document and its rarity across the corpus. Its objective is to reduce the impact of very common words while emphasizing terms that are more informative for distinguishing among classes.

The TF-IDF weight of a term t in a document d is computed as

$$TF-IDF(t, d) = TF(t, d) \times IDF(t)$$

where

$$IDF(t) = \log \left(\frac{N}{df(t)} \right)$$

with N representing the total number of documents and $df(t)$ the number of documents containing term t .

This weighting scheme reduces the influence of frequent but weakly informative terms and often improves classification performance compared to raw

frequency counts alone.

4.1.2 Dense Word Embeddings

Although frequency-based approaches rely primarily on lexical occurrence patterns, dense word embeddings aim to capture semantic relationships between words.

Word embeddings are continuous vector representations learned from text corpora according to the distributional hypothesis, namely the idea that words occurring in similar contexts tend to have similar meanings. In this study, embeddings are generated using the Word2Vec framework.

Two Word2Vec architectures are considered. The Continuous Bag of Words (CBOW) model predicts a target word from its surrounding context, whereas the Skip-Gram model predicts surrounding words given a target word. Although both approaches learn semantic representations, they exhibit different characteristics.

CBOW tends to produce smoother and more stable embeddings because it aggregates contextual information before making predictions. This architecture is generally computationally efficient and performs well when training data is limited. In the present experiments, CBOW embeddings are trained with 50 dimensions.

Conversely, Skip-Gram learns by predicting context words individually and often captures more fine-grained semantic relationships, particularly for infrequent terms. Because of its ability to model rare words more effectively, Skip-Gram is often preferred when semantic expressiveness is prioritized over training speed. In this study, Skip-Gram embeddings are trained with 100 dimensions.

Since classification is performed at the document level rather than the word level, individual word embeddings are aggregated through mean pooling. Given a document containing n words with embeddings $\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_n$, the document representation is computed as

$$\mathbf{d} = \frac{1}{n} \sum_{i=1}^n \mathbf{w}_i$$

This strategy provides a fixed-length document representation while maintaining a relatively low computational cost.

4.1.3 Transformer-Based Representations

Recent advances in Natural Language Processing have demonstrated the effectiveness of transformer architectures for learning contextualized textual representations.

Unlike BoW, TF-IDF, and Word2Vec, transformer encoders generate token representations that depend on the entire input sequence. Consequently, the meaning assigned to a word varies according to its surrounding context, allowing the model to capture lexical ambiguity, long-range dependencies, and complex semantic interactions.

The transformer architecture is based on the self-attention mechanism, which computes relationships among all tokens within a sequence. Given query Q , key K , and value V matrices, attention is defined as

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

where d_k denotes the dimensionality of the key vectors.

Two transformer architectures are considered in the analysis:

- **BERT** (*bert-base-uncased*);
- **RoBERTa** (*roberta-base*).

BERT (Bidirectional Encoder Representations from Transformers) is pre-trained through masked language modeling and next-sentence prediction objec-

tives. Its bidirectional training strategy allows it to exploit both left and right context simultaneously, making it effective for contextual language understanding.

RoBERTa (Robustly Optimized BERT Approach) extends BERT by introducing improvements in the pre-training procedure, including larger training corpora, dynamic masking strategies, and the removal of the next-sentence prediction objective. These modifications generally yield stronger language representations and improved downstream performance.

Rather than using transformer encoders only as static feature extractors, both models are fine-tuned directly on the privacy-risk classification task, enabling the learned representations to adapt to the linguistic and semantic characteristics of the target datasets.

4.2 Classification Models

To evaluate privacy-risk classification under different learning paradigms, five supervised machine learning algorithms are employed: Logistic Regression (LR), Support Vector Machine (SVM), Multinomial Naive Bayes (MNB), Decision Tree (DT), and Random Forest (RF). These models were selected because they represent complementary approaches to classification, ranging from linear and probabilistic methods to non-linear tree-based and ensemble techniques. Their inclusion makes it possible to compare simple and interpretable baselines with more flexible decision models.

Logistic Regression is a discriminative linear classifier that models the posterior probability of a class through the logistic function. Given an input feature vector $\mathbf{x}$, the probability of belonging to the positive class is defined as

$$P(y = 1|\mathbf{x}) = \frac{1}{1 + e^{-(\mathbf{w}^T \mathbf{x} + b)}}$$

where $\mathbf{w}$ denotes the vector of learned coefficients and b represents the in-

tercept term.

Although conceptually simple, Logistic Regression is particularly effective for text classification because textual representations are typically high-dimensional and sparse. In such settings, linear decision boundaries often provide strong predictive performance while remaining computationally efficient and relatively easy to interpret. Another important advantage is the possibility of controlling model complexity through regularization. L1 regularization promotes sparse solutions by forcing some coefficients to zero, L2 regularization penalizes large coefficients and improves stability, and Elastic Net combines both penalties. These mechanisms help reduce overfitting and improve generalization, especially when the number of features is very large compared to the number of training instances.

Support Vector Machine is another powerful classifier for textual data. Its objective is to identify the decision boundary that maximizes the margin between classes. Formally, in the hard-margin case, SVM solves the optimization problem

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2$$

subject to

$$y_i(\mathbf{w}^T \mathbf{x}_i + b) \geq 1$$

for every training example.

The margin-maximization principle generally leads to strong generalization capabilities, which explains why SVM has historically been one of the most successful traditional approaches for text classification. In high-dimensional spaces, the decision boundary is determined by a subset of training examples known as support vectors. In practice, both linear and non-linear kernels can be adopted. The linear kernel is particularly appropriate for sparse textual repre-

sentations because of its efficiency and scalability, while RBF, polynomial, and sigmoid kernels provide more flexible non-linear decision surfaces. The regularization parameter C controls the trade-off between maximizing the margin and minimizing classification errors, whereas for non-linear kernels the parameter γ influences the locality and complexity of the decision function.

Multinomial Naive Bayes is a probabilistic classifier derived from Bayes' theorem:

$$P(c|d) = \frac{P(d|c)P(c)}{P(d)}$$

where c denotes a class and d represents a document.

The model assumes conditional independence among features given the class label. Although this assumption is clearly simplified and rarely holds exactly in natural language, Multinomial Naive Bayes often performs remarkably well in text classification because word frequencies tend to provide strong class-specific signals. The multinomial formulation is specifically designed for count-based textual features and therefore naturally fits BoW and TF-IDF representations. During training, the model estimates class-specific term distributions, and prediction is performed by assigning the class associated with the highest posterior probability. The main hyperparameter is the smoothing coefficient α , which prevents zero probabilities for unseen terms and improves robustness, especially when the training data is limited. Because Multinomial Naive Bayes assumes non-negative count-based features, it is not suitable for dense Word2Vec representations.

Decision Tree is a non-linear classifier that predicts the class label through a hierarchy of feature-based decision rules. At each node, the algorithm selects the feature that maximizes class separation according to an impurity criterion. One of the most commonly used criteria is Gini impurity:

$$G = 1 - \sum_{i=1}^K p_i^2$$

where p_i represents the proportion of samples belonging to class i .

The recursive partitioning process continues until predefined stopping conditions are met. A major advantage of Decision Trees lies in interpretability, since each prediction can be traced through an explicit sequence of branching decisions. However, this flexibility also makes them prone to overfitting, particularly in sparse and high-dimensional text classification settings where many features may encode noisy or weakly informative patterns. For this reason, hyperparameters such as maximum depth, minimum samples per split, minimum samples per leaf, and splitting criterion are tuned to limit model complexity and improve generalization.

Random Forest extends the Decision Tree paradigm through ensemble learning. Instead of relying on a single tree, the algorithm builds multiple decision trees using bootstrap samples of the training data and random subsets of features. Each tree therefore learns a slightly different view of the data, and their predictions are aggregated to produce a more stable final decision.

The final prediction is obtained through majority voting:

$$\hat{y} = \text{mode}(T_1(\mathbf{x}), T_2(\mathbf{x}), \dots, T_n(\mathbf{x}))$$

where T_i denotes the prediction of the i -th tree.

By combining multiple weakly correlated trees, Random Forest usually reduces the variance of a single Decision Tree and improves robustness. This property often makes it less sensitive to overfitting than an individual tree. However, its effectiveness in textual classification is not always guaranteed, because sparse high-dimensional feature spaces may limit the quality of tree-based splits. The most relevant hyperparameters include the number of estimators, maximum tree depth, minimum samples required for splitting, minimum samples per leaf, and the splitting criterion.

4.3 Experimental Workflow

For all datasets except SPeDaC, the available data are divided into training and testing subsets using a stratified 70%/30% split. Stratification ensures that the original class distribution is preserved in both partitions, reducing potential biases during evaluation.

For the SPeDaC dataset, the official Train/Validation/Test partition released with the benchmark is retained. Using the original split guarantees comparability with previous studies and preserves the experimental protocol proposed by the dataset creators.

Unless otherwise specified, textual preprocessing includes lowercasing, tokenization, stopword removal, and basic normalization. The same preprocessing pipeline is applied consistently across all frequency-based representations to ensure fair experimental conditions.

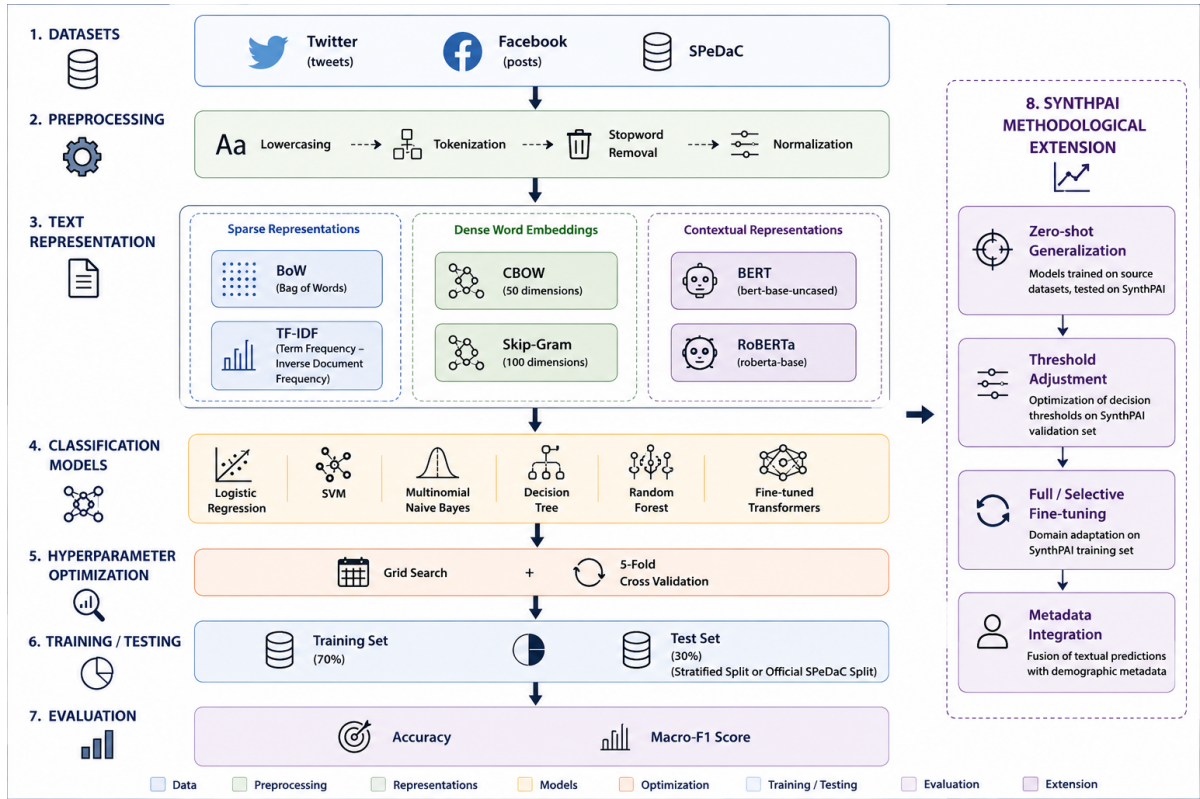


Figure 4.1: Overview of the experimental framework adopted in this study for privacy-risk classification.

The overall experimental framework adopted in this study is summarized in Figure 4.1. The workflow illustrates the complete experimental procedure, including dataset preparation, preprocessing, textual representation, model training, hyperparameter optimization, evaluation, and the additional methodological extension performed on the SynthPAI dataset.

Table 4.1: Class distribution of the datasets used in the experiments.

Dataset	Split	Non-private (0)	Private (1)	Total
Twitter (Cappellari)	Train	3091	1051	4142
Twitter (Cappellari)	Test	1326	450	1776
Facebook (Bioglio-Pensa)	Train	1742	1011	2753
Facebook (Bioglio-Pensa)	Test	748	433	1181
SPeDaC	Train	5325	5359	10684
SPeDaC	Validation	404	442	846
SPeDaC	Test	1132	1087	2219

Table 4.1 reports the distribution of class labels across the datasets employed in the experiments. Twitter and Facebook exhibit a moderate imbalance toward the non-private class, whereas SPeDaC is substantially balanced between sensitive and non-sensitive instances. The use of stratified splits for Twitter and Facebook preserves the original label proportions in the training and test partitions, while the official SPeDaC partition is maintained to ensure comparability with previous studies.

4.4 Transformer Fine-Tuning

To exploit contextual language representations, BERT and RoBERTa are fine-tuned directly on the privacy-risk classification task. For both models, a linear classification head is added on top of the transformer encoder. Fine-tuning is performed for 3 epochs using the AdamW optimizer with a learning rate of 2×10^{-5} and a maximum sequence length of 128 tokens. AdamW combines adaptive gradient estimation with weight decay regularization, allowing more stable optimization and improved generalization. Dropout regularization is also applied during training to reduce overfitting.

Given the relatively short training horizon, early stopping is not employed. Nevertheless, validation performance is monitored throughout training to verify convergence stability and identify potential optimization issues.

4.5 Hyperparameter Optimization

The selection of appropriate hyperparameters is crucial for obtaining reliable and reproducible results. Consequently, model selection is performed through grid search combined with 5-fold cross-validation on the training set.

In 5-fold cross-validation, the training data are divided into five equally sized subsets. During each iteration, four folds are used for training and one fold is used for validation. The process is repeated five times, allowing each

fold to serve as a validation set exactly once. The final score corresponds to the average performance across all folds.

Cross-validated accuracy is used as the model selection criterion. The hyperparameter configuration achieving the highest average validation score is subsequently retrained on the complete training set before final evaluation.

The explored search space includes:

- **Logistic Regression:** penalty type, solver, and *l1_ratio* for Elastic Net regularization;
- **Support Vector Machine:** kernel type and regularization settings;
- **Multinomial Naive Bayes:** smoothing parameter α ;
- **Decision Tree:** maximum depth, split criterion, minimum samples split, and minimum samples leaf;
- **Random Forest:** number of estimators, maximum depth, split criterion, minimum samples split, and minimum samples leaf.

Multinomial Naive Bayes is excluded from CBOW and Skip-Gram experiments because its probabilistic formulation assumes count-based non-negative features and is therefore unsuitable for dense continuous embeddings.

4.6 Evaluation Metrics

The final evaluation is conducted on the held-out test set using Accuracy and Macro-F1 score.

Accuracy measures the proportion of correctly classified instances among all predictions:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

where TP denotes true positives, TN true negatives, FP false positives, and FN false negatives.

Although accuracy provides an intuitive measure of overall predictive performance, it can be misleading when class distributions are imbalanced.

To obtain a more balanced evaluation, Precision and Recall are first computed as

$$Precision = \frac{TP}{TP + FP}$$

and

$$Recall = \frac{TP}{TP + FN}$$

The F1-score is then calculated as the harmonic mean of Precision and Recall:

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

For multi-class classification, the F1-score is computed independently for each class and subsequently averaged:

$$MacroF1 = \frac{1}{K} \sum_{i=1}^K F1_i$$

where K denotes the number of classes.

Macro-F1 assigns equal importance to all classes regardless of their frequency and therefore provides a more informative assessment when class imbalance is present. The combined use of Accuracy and Macro-F1 enables a comprehensive evaluation of both overall predictive effectiveness and class-wise robustness.

4.7 Methodological Extension: Evaluation on SynthPAI

In addition to the experiments conducted on the primary privacy-risk datasets, a further methodological extension was carried out on the SynthPAI dataset. The purpose of this analysis was not limited to measuring predictive performance on an additional corpus, but also included the investigation of three broader issues: cross-dataset generalization, domain adaptation, and the potential contribution of auxiliary demographic information.

SynthPAI differs from the previously considered datasets in several respects, including textual style, profile structure, and the availability of user-level metadata. For this reason, it provides a useful testbed for examining whether privacy-risk classifiers trained in one setting can transfer effectively to a different environment, and whether adaptation strategies can mitigate the resulting domain shift.

4.7.1 Construction of the SynthPAI Evaluation Setting

The dataset was first preprocessed in order to produce a clean document-level collection suitable for classification. Unused structural fields, such as thread identifiers and parent-child relations, were removed. In addition, profile information associated with each record was parsed in order to extract a set of structured user attributes.

Since SynthPAI does not provide the same privacy-risk labels used in the primary experiments, a binary proxy label was constructed from the available annotation structure. In particular, a post was considered privacy-relevant when the associated guesses field was not empty, and non-risk otherwise. This operationalization allowed the dataset to be reformulated as a binary classification task, making it compatible with the previously trained models and with the fine-tuning procedures adopted in the subsequent experiments.

Beyond the textual content, additional user-level variables were derived from

the profile metadata. These include age, age group, gender, occupational macro-category, education level, and a mobility indicator based on the comparison between current and birth location. Such attributes were not used in the initial cross-dataset experiments, but were later incorporated into a hybrid text-plus-metadata setting.

Profile Attribute Normalization and Aggregation

In the construction of the SynthPAI evaluation setting, profile metadata was not used exclusively in their original raw form. Instead, selected attributes were transformed into a reduced set of derived variables to obtain a more consistent and analytically interpretable representation of user characteristics. This pre-processing step was motivated by the heterogeneous and highly granular nature of several profile fields, which would otherwise make comparison between users more difficult and introduce excessive sparsity in the analysis.

More specifically, age was extracted from the profile metadata and converted into the categorical variable age range, using four intervals: 18-24, 25-39, 40-59, and 60+. This discretization made it possible to compare users at a broader demographic level rather than relying on exact numerical ages. The gender was derived from the original sex field and binarized according to the annotation format available in the dataset.

Occupation was also normalized through a mapping from fine-grained job titles to a smaller set of macro-categories, namely *Creative*, *STEM/Tech*, *Education/Culture*, *Healthcare*, *Business*, and *Other*. This aggregation was introduced to reduce lexical variation and to make occupational information more robust for descriptive and comparative purposes. A similar transformation was applied to education, where heterogeneous textual labels were grouped into broader educational levels such as *High School*, *Undergraduate*, *Graduate*, *Post-Graduate*, and *Other*.

Finally, a mobility-related variable was derived by comparing the user’s current location with the reported birthplace. When the values of `city_country`

and `birth_city_country` differed, the user was marked geographically mobile; otherwise, the profile was treated geographically stable. Although simplified, this variable provides a compact proxy for spatial mobility based on the meta-data available in the dataset.

These derived variables should be interpreted as analytical abstractions introduced for methodological consistency, rather than as exhaustive representations of personal identity. Their purpose is not to add new information, but to operationalize existing profile metadata into a form better suited for structured analysis within the SynthPAI-based evaluation setting.

Table 4.2: Derived variables used in the SynthPAI evaluation setting.

Original field	Derived variable	Transformation
<code>age</code>	<code>age_range</code>	grouped into 18–24, 25–39, 40–59, and 60+
<code>sex</code>	<code>gender</code>	binarized according to available labels
<code>occupation</code>	<code>job_macro</code>	mapped to broad occupational domains
<code>education</code>	<code>edu_level</code>	normalized into macro educational levels
<code>city_country</code> , <code>birth_city_country</code>	<code>is_mobile</code>	1 if current and birth location differ, 0 otherwise

4.7.2 Cross-Dataset Generalization Protocol

The first methodological objective was to evaluate the extent to which privacy-risk classifiers trained on previous datasets could generalize to SynthPAI without any further adaptation.

To this end, two previously trained models were applied directly to the dataset:

- a **Logistic Regression** classifier based on TF-IDF representations;
- a **RoBERTa** classifier fine-tuned on one of the original privacy-risk datasets.

In both cases, the models were used in a zero-shot configuration, which means that no additional training was performed on SynthPAI before evaluation.

This setup provides a direct measure of cross-dataset transferability and makes it possible to assess the effects of domain shift on both traditional and transformer-based approaches.

The comparison between Logistic Regression and RoBERTa also supports a broader methodological question. Traditional models depend strongly on lexical overlap between training and test corpora, whereas transformer models encode richer contextual information that may, in principle, generalize more effectively across domains. Evaluating both families of models on SynthPAI therefore makes it possible to test whether contextualized representations are more robust than sparse lexical features under dataset shift.

4.7.3 Decision Threshold Adjustment

Because privacy-risk detection is a safety-sensitive classification task, particular attention was devoted to the trade-off between false positives and false negatives. In many practical scenarios, failing to detect a potentially sensitive disclosure may be more problematic than incorrectly flagging a benign post.

For this reason, the RoBERTa model was also evaluated under a modified decision rule. Instead of assigning the positive label only when the predicted probability exceeded the conventional threshold of 0.5, a lower threshold of 0.2 was also tested. From a methodological perspective, this experiment makes it possible to study whether model sensitivity can be increased without changing the underlying parameters, and to assess the resulting trade-off between precision and recall.

4.7.4 Domain Adaptation through Fine-Tuning

To investigate whether the performance degradation caused by domain shift could be mitigated, RoBERTa was further adapted to SynthPAI through supervised fine-tuning.

For this purpose, the dataset was divided into training and test subsets

using an 80%/20% split. The textual inputs were tokenized with the same tokenizer associated with the pre-trained RoBERTa checkpoint, using truncation and padding up to a maximum sequence length of 128 tokens. Fine-tuning was then carried out on the SynthPAI training subset using the Hugging Face training framework.

This stage allows the model parameters to specialize toward the linguistic and semantic characteristics of SynthPAI, rather than relying exclusively on representations learned from external corpora. The resulting experiment provides a direct estimate of the benefit introduced by in-domain adaptation.

4.7.5 Selective Fine-Tuning

In addition to full fine-tuning, a selective fine-tuning strategy was explored in order to reduce computational cost and limit the number of trainable parameters. In this configuration, the embedding layer and the lower transformer layers were frozen, while only the upper layers and the classification head were updated. The methodological motivation for this choice lies in the hierarchical organization of transformer representations: lower layers tend to capture more general syntactic and lexical information, whereas higher layers are more strongly associated with task-specific abstractions.

Selective fine-tuning therefore makes it possible to test whether acceptable adaptation performance can be achieved while preserving most of the original pre-trained representations and reducing optimization complexity.

4.7.6 Integration of Demographic Metadata

A final methodological extension concerned the integration of structured user-level information into the classification pipeline.

After fine-tuning RoBERTa on SynthPAI, the model’s output probabilities were extracted for each post. These probability scores were then combined with selected demographic attributes, including gender, age group, and occupational

macro-category. Categorical features were converted into binary indicator variables through one-hot encoding, and the resulting feature set was used as input to an XGBoost classifier.

This hybrid configuration was designed to test whether privacy-risk prediction can benefit from the joint use of textual and demographic information. From a methodological perspective, the experiment makes it possible to examine whether structured metadata provide complementary predictive value beyond what is already captured by the contextual language model.

4.7.7 Research Questions

Overall, the methodological extension on SynthPAI was designed to address three main research questions:

1. To what extent can privacy-risk classifiers generalize across datasets without further adaptation?
2. Does fine-tuning on the target corpus improve predictive performance under domain shift?
3. Can demographic metadata provide additional predictive value when combined with text-based representations?

Answering these questions contributes to a broader understanding of the conditions under which privacy-risk classification models remain robust, adaptable, and informative across heterogeneous data environments.

Chapter 5

Experimental Evaluation

This chapter presents the empirical results obtained on the selected datasets. It first reports the outcomes of hyperparameter optimization and then discusses the final classification performance across datasets, text representations and model.

5.1 Hyperparameter Search Results

Hyperparameter optimization is performed through a grid search with a 5-fold cross-validation on the training portion of each dataset. Tables 5.1–5.3 summarize the best configurations identified for the Twitter, SPeDaC, and Facebook datasets, respectively.

Overall, the optimal settings vary across datasets and feature representations, confirming that model performance is strongly data-dependent. Among the traditional approaches, Support Vector Machine and Logistic Regression emerge as the most stable models across sparse representations, whereas Random Forest and Decision Tree show greater variability. Word2Vec-based representations produce less consistent results, especially in social media data, where textual brevity and lexical sparsity remain challenging factors.

Best configurations identified by grid search include the following:

- **SVM:** the optimal kernel varies across datasets and representations, including *poly*, *rbf*, and *linear*.
- **Logistic Regression:** *saga* with *elasticnet* on Twitter and *newton-cg* on SPeDaC.

- **Naive Bayes:**

$$\alpha = 3.0$$

for Twitter and Facebook, and

$$\alpha = 1.5$$

for SPeDaC.

- **Random Forest:** deeper configurations with a higher number of estimators are generally preferred on SPeDaC and Facebook.

Table 5.1: Hyperparameter optimization on the Twitter dataset (5-fold CV).

Technique	Model	Best Hyperparameters	CV Acc.
BoW	SVM	Kernel: Poly	0.746
	MNB	Alpha: 3.0	0.712
	RF	Depth: 20, Est: 10, Leaf: 5	0.746
	LR	SAGA, ElasticNet, 0.5	0.732
	DT	Entropy, Depth: 2, Leaf: 15	0.745
TF-IDF	SVM	Kernel: RBF	0.747
	MNB	Alpha: 2.0	0.746
	RF	Depth: 20, Est: 10, Leaf: 5	0.746
	LR	L2, LBFGS	0.745
	DT	Gini, Depth: 2, Leaf: 5	0.745
CBOW	SVM	Kernel: Linear	0.746
	RF	Entropy, Depth: 20, Est: 150	0.746
	LR	SAGA, ElasticNet, 0.5	0.746
	DT	Entropy, Depth: 2, Leaf: 15	0.746
Skip-Gram	SVM	Kernel: Linear	0.746
	RF	Gini, Depth: 10, Est: 50	0.747
	LR	SAGA, ElasticNet, 0.5	0.746
	DT	Entropy, Depth: 2, Leaf: 5	0.746

Table 5.2: Hyperparameter optimization on the SPeDaC dataset (5-fold CV).

Technique	Model	Best Hyperparameters	CV Acc.
BoW	SVM	Kernel: RBF	0.825
	MNB	Alpha: 1.5	0.813
	RF	Depth: 20, Est: 150, Leaf: 5	0.744
	LR	L2, LBFGS	0.832
	DT	Depth: 20, Leaf: 5	0.607
TF-IDF	SVM	Kernel: RBF	0.850
	MNB	Alpha: 1.5	0.810
	RF	Entropy, Depth: 20, Est: 150	0.756
	LR	Newton-CG	0.823
	DT	Depth: 20, Leaf: 20	0.619
CBOW	SVM	Kernel: Linear	0.608
	RF	Depth: 20, Est: 150, Leaf: 5	0.738
	LR	Newton-CG	0.636
	DT	Entropy, Depth: 20, Leaf: 5	0.655
Skip-Gram	SVM	Kernel: RBF	0.648
	RF	Entropy, Depth: 20, Est: 150	0.762
	LR	Newton-CG	0.685
	DT	Entropy, Depth: 20, Leaf: 5	0.685

Table 5.3: Hyperparameter optimization on the Facebook (Bioglio–Pensa) dataset (5-fold CV).

Technique	Model	Best Hyperparameters	CV Acc.
BoW	SVM	Kernel: RBF	0.714
	MNB	Alpha: 3.0	0.718
	RF	Gini, Depth: 20, Est: 10	0.655
	LR	L2, LBFGS	0.728
	DT	Entropy, Depth: 10, Leaf: 5	0.698
TF-IDF	SVM	Kernel: Linear	0.724
	MNB	Alpha: 1.5	0.693
	RF	Entropy, Depth: 20, Est: 20	0.664
	LR	L2, LBFGS	0.717
	DT	Entropy, Depth: 5, Leaf: 5	0.692
CBOW	SVM	Kernel: Poly	0.662
	RF	Entropy, Depth: 20, Est: 150	0.677
	LR	Newton-CG	0.651
	DT	Entropy, Depth: 2, Leaf: 5	0.620
Skip-Gram	SVM	Kernel: Linear	0.633
	RF	Entropy, Depth: 20, Est: 100	0.667
	LR	Newton-CG	0.661
	DT	Gini, Depth: 2, Leaf: 5	0.655

5.2 Classification Results

Table 5.4 reports the final performance of the test-set across text representations and model families on the three datasets. The results are presented in terms of accuracy and macro-F1 in order to capture both overall predictive performance and balance between classes.

Although CBOW and Skip-Gram representations were explored during the experimental phase, they are not included in the final comparative table because they generally underperformed with respect to sparse representations and transformer-based models across the evaluated datasets.

Table 5.4: Classification performance across text representations and models.

Text Rep.	Model	Twitter (Cappellari)		Facebook (Bioglio-Pensa)		SPeDaC	
		Acc.	F1	Acc.	F1	Acc.	F1
BoW	SVM	0.746	0.639	0.718	0.689	0.981	0.981
	MNB	0.745	0.645	0.718	0.708	0.937	0.937
	RF	0.746	0.638	0.651	0.561	0.789	0.782
	LR	0.728	0.677	0.735	0.727	0.983	0.983
	DT	0.743	0.643	0.690	0.647	0.641	0.596
TF-IDF	SVM	0.745	0.642	0.716	0.708	0.998	0.998
	MNB	0.746	0.638	0.683	0.621	0.935	0.936
	RF	0.746	0.638	0.650	0.561	0.812	0.815
	LR	0.745	0.638	0.715	0.692	0.998	0.999
	DT	0.744	0.642	0.681	0.662	0.655	0.619
Transformer	BERT	0.720	0.412	0.832	0.774	0.992	0.993
	RoBERTa	0.711	0.415	0.842	0.796	0.987	0.988

In the Twitter dataset, traditional sparse representations provide the most stable and competitive results. BoW and TF-IDF combined with SVM, Logistic Regression, and Multinomial Naive Bayes achieve similar performance levels, with accuracy values around 0.74–0.75 and macro-F1 scores generally between 0.63 and 0.68. Among these models, Logistic Regression with BoW attains the highest macro-F1, suggesting that relatively simple lexical representations remain highly informative for short and noisy social media texts.

By contrast, transformer-based models do not outperform the best classical baselines in this dataset. Although BERT and RoBERTa obtain acceptable accuracy values, their macro-F1 scores are substantially lower, indicating weaker balance between classes. A plausible explanation is that the Twitter corpus is characterized by short length, high linguistic variability, and limited contextual information, which may reduce the benefits of fine-tuning large pre-trained language models.

In the Facebook dataset, transformer-based models obtain the strongest results. RoBERTa achieves the best overall performance, with accuracy followed by BERT. This suggests that contextualized language representations are particularly beneficial in this setting, likely because Facebook posts are generally longer and semantically richer than tweets, allowing transformer models to ex-

exploit broader contextual dependencies.

Among the traditional approaches, Logistic Regression and Multinomial Naive Bayes remain reasonably competitive, especially with BoW features, whereas tree-based methods perform less effectively. Random Forest and Decision Tree show lower macro-F1 values, confirming the difficulty of handling high-dimensional sparse textual data through recursive partitioning alone.

The SPeDaC dataset appears considerably more separable than the social media corpora. Both BoW and TF-IDF combined with linear models achieve near-ceiling performance, with Logistic Regression and SVM reaching accuracy and macro-F1 values close to 1.0. In particular, TF-IDF with Logistic Regression and SVM yields the strongest results, showing that sparse lexical representations are already sufficient to capture most of the discriminative information contained in the dataset.

Transformer-based models also perform extremely well on SPeDaC, but they do not provide substantial gains over the best traditional baselines. This suggests that the classification task is already highly tractable using simpler models and that the classes are characterized by strong lexical regularities.

Overall, the results reveal a strong dataset-dependent pattern. Traditional sparse features paired with linear classifiers form a strong and reliable baseline on Twitter and, especially, on SPeDaC, where SVM and Logistic Regression approach ceiling performance. On Facebook, on the contrary, transformer models show a clear advantage, suggesting that richer contextual information becomes more valuable when texts are longer and more semantically articulated.

These findings indicate that annotation choices, privacy definitions, writing style, and contextual structure materially shape model behavior and constrain cross-dataset generalization. More broadly, they reinforce the need for more standardized, diverse, and interoperable resources for privacy-focused natural language processing research.

To complement the aggregate performance metrics reported in Table 5.4, Table 5.5 presents the distribution of predicted labels across datasets, text rep-

representations and models. In particular, the table reports the number of instances assigned to class 0 and class 1 by each classifier. This analysis is useful for identifying degenerate behaviors, such as collapse toward the majority class, which may yield apparently acceptable accuracy values despite poor recognition of the minority class.

Table 5.5: Predicted class distribution across datasets, text representations, and models.

Dataset	Representation	Model	Pred. Class 0	Pred. Class 1
Twitter (Cappellari)	BoW	SVM	1773	3
	BoW	DT	1758	18
	BoW	RF	1776	0
	BoW	MNB	1760	16
	BoW	LR	1614	162
	TF-IDF	SVM	1766	10
	TF-IDF	DT	1762	14
	TF-IDF	RF	1776	0
	TF-IDF	MNB	1776	0
	TF-IDF	LR	1775	1
	CBOW	SVM	1776	0
	CBOW	DT	1776	0
	CBOW	RF	1776	0
	CBOW	LR	1776	0
	Skip-Gram	SVM	1776	0
	Skip-Gram	DT	1776	0
	Skip-Gram	RF	1754	22
	Skip-Gram	LR	1776	0
SPeDaC	BoW	SVM	1102	1117
	BoW	DT	1847	372
	BoW	RF	1115	1104
	BoW	MNB	1057	1162
	BoW	LR	1113	1106
	TF-IDF	SVM	1134	1085
	TF-IDF	DT	1789	430
	TF-IDF	RF	1190	1029
	TF-IDF	MNB	1053	1166
	TF-IDF	LR	1133	1086
	CBOW	SVM	1347	872
	CBOW	DT	1167	1052
	CBOW	RF	1134	1085
	CBOW	LR	1159	1060
	Skip-Gram	SVM	1373	846
	Skip-Gram	DT	1148	1071
	Skip-Gram	RF	1140	1079
	Skip-Gram	LR	1167	1052
Facebook (Bioglio-Pensa)	BoW	SVM	963	218
	BoW	DT	990	191
	BoW	RF	1131	50
	BoW	MNB	856	325
	BoW	LR	850	331
	TF-IDF	SVM	839	342
	TF-IDF	DT	904	277
	TF-IDF	RF	1094	87
	TF-IDF	MNB	1062	119
	TF-IDF	LR	938	243
	CBOW	SVM	1036	145
	CBOW	DT	674	507
	CBOW	RF	964	217
	CBOW	LR	944	237
	Skip-Gram	SVM	1181	0
	Skip-Gram	DT	1013	168
	Skip-Gram	RF	969	212
	Skip-Gram	LR	889	292

Table 5.5 reveals substantial differences in class-wise prediction behavior between datasets. On Twitter (Cappellari), many models assign almost all instances to class 0, indicating a marked collapse toward the majority class. This behavior helps explain why several configurations achieve apparently acceptable accuracy values while exhibiting weak macro-F1 and extremely poor recognition of the positive class. In these cases, the models are not effectively separating the two categories, but are instead exploiting the imbalanced class distribution.

This pattern is considerably less evident on SPeDaC, where predicted labels remain much more balanced across most representations and models. The result is consistent with the much higher performance observed on this dataset and suggests that the two classes are more clearly separable. Facebook (Bioglio-Pensa) presents an intermediate scenario: some classifiers remain relatively balanced, especially under sparse representations, whereas others still show a clear preference for class 0.

The strong asymmetry observed in several Twitter configurations should not be interpreted as a technical malfunction of the classifiers. Rather, it reflects a well-known phenomenon in imbalanced classification settings, where models optimized primarily for accuracy may converge toward majority-class predictions. For this reason, macro-F1 and class-sensitive analyzes are more informative than accuracy alone when evaluating privacy-risk classification systems.

5.3 Evaluation on SynthPAI

This section reports the results of the experiments conducted on the SynthPAI dataset. The evaluation is organized into five parts: zero-shot cross-dataset generalization, decision threshold analysis, full fine-tuning, selective fine-tuning, and integration of demographic metadata.

Table 5.6 summarizes the macro-F1 scores obtained in all experimental configurations.

Table 5.6: Summary of results on SynthPAI across experimental configurations.

Configuration	Accuracy	Precision	Recall	Macro-F1
Logistic Regression (zero-shot)	0.50	0.50	0.50	0.50
RoBERTa (zero-shot, threshold 0.5)	0.55	0.60	0.51	0.39
RoBERTa (zero-shot, threshold 0.2)	0.57	0.61	0.53	0.43
RoBERTa (full fine-tuning)	0.72	0.72	0.71	0.71
RoBERTa (selective fine-tuning)	0.69	0.69	0.69	0.69
RoBERTa + XGBoost + metadata	0.72	0.72	0.72	0.72

5.3.1 Zero-Shot Cross-Dataset Generalization

The first experiment evaluates the generalization capability of classifiers trained on previous datasets when applied directly to SynthPAI without further adaptation.

The Logistic Regression model trained with TF-IDF features achieves an accuracy of 0.50 and a macro-F1 of 0.50, effectively performing at chance level. The confusion matrix confirms this result: the model produces a nearly balanced split between true negatives (2125) and false positives (2148) for the non-risk class, and between false negatives (1783) and true positives (1767) for the risk class. The per-class F1-scores are 0.52 and 0.47 for the negative and positive classes, respectively, indicating that no meaningful discriminative pattern is transferred from the training domain.

The RoBERTa model applied with the standard threshold of 0.5 achieves a slightly higher accuracy of 0.55, but produces a strongly imbalanced prediction pattern. While recall for the negative class reaches 0.98, recall for the positive class collapses to 0.04, yielding a macro-F1 of only 0.39. This result reveals a severe prediction bias towards the negative class, suggesting that the model systematically underestimates the risk of privacy outside of its training distribution.

These results confirm that both lexical and contextual models suffer from significant performance degradation under domain shift, and that cross-dataset generalization for privacy-risk detection remains a challenging open problem.

5.3.2 Decision Threshold Analysis

To address the strong bias toward the negative class observed in the zero-shot RoBERTa configuration, the classification threshold was lowered from 0.5 to 0.2. The rationale for this adjustment is that privacy-risk detection can be considered a recall-sensitive task, in which failure to identify a risky disclosure may be more consequential than generating a false alarm.

With a threshold of 0.2, accuracy increases marginally to 0.57, while the recall for the positive class improves from 0.04 to 0.09. However, the overall macro-F1 remains low at 0.43, and the precision for the positive class drops to 0.67. The improvement in recall is therefore modest, and the overall imbalance between classes persists.

These results indicate that threshold adjustment alone is insufficient to compensate for the domain gap between the training data and SynthPAI. Although a lower threshold increases the sensitivity of the model, it does not address the underlying distributional mismatch that limits generalization.

5.3.3 Full Fine-Tuning

To enable direct adaptation to the SynthPAI domain, the RoBERTa model was fine-tuned on a SynthPAI training subset of 5 epochs, using a learning rate of 2×10^{-5} and a batch size of 4 with accumulation of gradients over 4 steps.

After fine-tuning, performance improves substantially. On the held-out test set, the model achieves an accuracy of 0.72 and a macro-F1 of 0.71. Per-class results show a precision of 0.73 and recall of 0.75 for the negative class and a precision of 0.70 with recall of 0.67 for the positive class. These results represent a large improvement over zero-shot configurations and demonstrate that domain adaptation through fine-tuning is effective for privacy-risk classification on SynthPAI.

The gain observed after fine-tuning is consistent with the hypothesis that SynthPAI contains domain-specific linguistic patterns and disclosure behaviors

that are not captured by models trained on other corpora. Updating the model parameters using in-domain data allows the learned representations to specialize toward the characteristics of the target dataset.

5.3.4 Selective Fine-Tuning

As an alternative to full fine-tuning, a selective fine-tuning strategy was evaluated. In this configuration, the eight lower layers of the RoBERTa encoder and the embedding layer were frozen, while only the upper four upper layers of the transformer and the classification head were updated. Fine-tuning was performed for 2 epochs with a learning rate of 5×10^{-5} .

The rationale for this approach is grounded in the hierarchical structure of transformer representations. The lower layers tend to encode general linguistic knowledge, such as syntax and basic semantics, whereas the upper layers capture more task-specific information. Updating only the upper layers therefore aims to preserve the general language understanding acquired during pre-training while reducing computational requirements and limiting the risk of overfitting.

The selective fine-tuning configuration achieves an accuracy of 0.69 and a macro-F1 of 0.69, with per-class F1-scores of 0.72 and 0.66 for the negative and positive classes, respectively. Although these results are slightly below those obtained by complete fine-tuning, the difference in macro-F1 is modest (0.69 versus 0.71), suggesting that selective fine-tuning provides a reasonable trade-off between adaptation effectiveness and computational efficiency.

5.3.5 Integration of Demographic Metadata

The final experiment investigates whether the integration of user-level demographic information can improve privacy-risk detection beyond what is achievable through textual analysis alone.

The fine-tuned RoBERTa model was used to generate probability scores for each document. These scores were combined with demographic features

extracted from the SynthPAI profiles, including age group, gender, educational background, and occupational category. The resulting feature matrix was used to train an XGBoost classifier.

This hybrid architecture integrates semantic information derived from text with structured user-level attributes, with the goal of capturing both linguistic indicators of privacy risk and broader behavioral patterns associated with personal disclosure.

The combined model achieves an accuracy of 0.72 and a macro-F1 of 0.72, with per-class F1-scores of 0.75 for the negative class and 0.69 for the positive class. Compared to full fine-tuning alone, precision for the positive class improves marginally from 0.70 to 0.71, while recall decreases slightly from 0.67 to 0.67. The overall macro-F1 improves by one point with respect to the complete fine-tuning baseline.

The modest but consistent gain suggests that demographic features provide complementary information to textual content, although the improvement is limited. This finding may reflect the fact that privacy-risk disclosure is primarily driven by the content and phrasing of individual posts rather than by demographic characteristics alone.

5.3.6 Discussion

Table 5.6 provides a comparative overview of all the configurations evaluated on SynthPAI. Several observations can be drawn from the results.

First, zero-shot generalization is consistently poor for both traditional and transformer-based models. The Logistic Regression classifier performs at the chance level, while RoBERTa exhibits a strong prediction bias that is only partially mitigated by threshold adjustment. These results underscore the challenge of cross-dataset transfer in privacy-risk detection and highlight the strong dependence of model performance on domain-specific characteristics.

Second, fine-tuning on SynthPAI data produces substantial improvements,

with macro-F1 increasing from 0.39 to 0.71 for full fine-tuning. This confirms that domain adaptation is a necessary step when deploying privacy-risk classifiers across different datasets.

Third, selective fine-tuning achieves results close to those of full fine-tuning at a lower computational cost, representing a viable alternative when resources are constrained.

Finally, the integration of demographic metadata provides a marginal but consistent improvement over text-only approaches, suggesting that user-level attributes carry some complementary predictive signal. However, the limited magnitude of the gain indicates that textual content remains the primary source of information for classification of privacy-risk.

Chapter 6

Conclusions and Future Work

This chapter presents the conclusions reached in this study and then outlines potential future developments.

6.1 Conclusions

This thesis investigated the extent to which privacy risk detection models can generalize across heterogeneous social media datasets. The work was motivated by the growing difficulty of protecting personal information in online environments, where sensitive cues may emerge not only through explicit disclosures, but also through indirect textual patterns, contextual signals, and platform-specific communication practices. To address this problem, the thesis adopted a cross-dataset evaluation framework and compared different datasets, feature representations, and classification settings for privacy-sensitive content detection. This choice made it possible to assess not only the performance of the model within individual datasets, but also the extent to which the learned patterns remain effective when transferred across different social media contexts.

The results show that privacy-related signals are strongly influenced by dataset-specific characteristics, including annotation criteria, domain conventions, and linguistic variability. As a consequence, models that achieve promising results in within-dataset settings do not necessarily maintain the same level of performance in cross-dataset experiments. This finding suggests that a substantial portion of the predictive power observed in isolated benchmarks may depend on local regularities rather than on truly generalizable privacy-related features.

Another relevant outcome of this work concerns the heterogeneity of the datasets considered. The resources analyzed differ in terms of platform origin, operational definition of privacy, granularity of the annotation, and target phenomena. These differences make direct comparison difficult and highlight the methodological challenge of building unified evaluation settings for privacy-aware machine learning. At the same time, they confirm the importance of studying privacy detection beyond single-dataset scenarios, especially when the final goal is to design models that can operate robustly in realistic and variable environments.

In general, this thesis contributes to the literature in two main ways. First, it provides a comparative analysis of several datasets used for detecting privacy-related content on social media, highlighting their distinctive characteristics and limitations. Second, it offers an experimental assessment of cross-dataset generalization, showing that robustness across heterogeneous contexts remains an open challenge for current approaches. In this sense, the work supports the view that privacy-sensitive content detection should be evaluated not only in terms of classification accuracy but also in terms of transferability, consistency, and practical applicability across platforms.

6.2 Future Work

The initial objective of this thesis was to measure the risk of user privacy violation on social media at an operational level. Due to the limited availability of

real-world signals necessary to quantify exposure (propagation, network structure, user history), the thesis focused instead on content-centric proxies and cross-dataset evaluations to assess how well models detect privacy-relevant signals and generalize across platforms. Building on the results and the constraints encountered, we outline research directions to progress from proxy-based sensitivity detection toward actionable, data-based risk measurement and prevention.

6.2.1 Risk Score

The most evident conceptual limitation of the current work is that classifying a post as sensitive or non sensitive is not equivalent to measuring the risk that a user’s privacy will actually be violated. The two concepts are related but not interchangeable: a post disclosing health information published by a user with a handful of followers in a closed community poses a fundamentally different risk from the same disclosure made by a highly central node in a large public network, where resharing can amplify reach beyond any reasonable expectation. This distinction is absent from all the datasets evaluated in this thesis and from the majority of the literature on privacy-sensitive content classification. A more complete model of privacy risk should integrate at least two orthogonal dimensions.

The first is impact, which concerns the nature and severity of the disclosed information: the sensitivity of the GDPR category involved, the degree to which attributes can be inferred even without explicit personally identifiable information, the reversibility of the potential harm and the vulnerability of the individual concerned.

The second is exposure, which concerns how broadly and persistently the content can reach unintended audiences: the user’s follower count and network centrality, the historical virality of their posts, the probability of resharing into adjacent communities, and the persistence of the content in caches and third-party archives. The work of Alam [50] take a significant step in this direction with their Comprehensive Privacy Risk Scoring (CPRS) framework, which integrates user-attribute sensitivity, social graph structure measured via SimRank

similarity and PageRank centrality, and content-based entity-level analysis into a single composite score. Their central empirical finding is that methods that focus on any one of these dimensions in isolation content or graph or attributes systematically underestimate true risk; and that nearly 75% of users share personal information with privacy settings that leave them concretely vulnerable to inference and de-anonymization attacks [50].

The natural next step for the line of work initiated in this thesis is therefore to combine the content-detection capacity of the models evaluated here with the network and propagation signals needed to estimate exposure, and to design a composite risk index that is interpretable across specific regulatory and demographic contexts. The weights assigned to each dimension should not be universally fixed, but should reflect the normative priorities of the applicable legal framework (for instance, assigning a higher weight to health and political-orientation disclosures under GDPR Article 9) as well as the vulnerability profile of the target population, with stricter thresholds for minors or individuals belonging to protected groups. Validation of such an index is itself a non-trivial research problem and should combine human-agreement studies on perceived severity of disclosure scenarios with correlation analyzes against observable adverse outcomes such as account reports, content takedowns, or measurable reputational damage.

6.2.2 User-Level Cumulative Risk and Implicit Inference

A single post rarely constitutes a privacy violation in isolation; rather, risk accumulates as a user’s set of public contributions grows. Staab and collaborators [42] demonstrated this concretely with the PersonalReddit dataset: GPT-4 achieved approximately 85.5% top-1 accuracy in inferring sensitive attributes (location, age, occupation, income) from the aggregated Reddit histories of 520 users, even when no single comment contained an explicit identifier and even after commercial anonymization tools had removed all recognized named entities. The implication is that any system scoring posts independently, as the models evaluated in this thesis necessarily do, will systematically underestimate

the true risk borne by active users.

Addressing this limitation requires moving from post-level classifiers to user-level sequential models capable of:

- aggregating content over time to compute a cumulative exposure score that increases non-linearly as successive posts jointly enable attribute inference;
- detecting implicit inference signals such as linguistic style, cultural references, topic co-occurrence patterns, that allow adversaries to deduce sensitive attributes without explicit PII, as modeled in the SynthPAI dataset through persona-consistent conversational generation [47];
- tracking a privacy depletion metric that quantifies how much anonymity margin remains after each new disclosure and that flags high-risk temporal trajectories before a critical threshold is crossed.

6.2.3 Richer Classification: Multi-Label, Graded Sensitivity, and Uncertainty

The binary or multi-class formulations used in the datasets examined in this thesis are a simplification that the literature has already begun to question. SPeDaC [27] introduced a three-level hierarchy (binary, five macro-categories, 61 fine-grained categories) precisely to show that fine-grained classification is substantially harder: even advanced transformer architectures experienced significant performance degradation on the 61-class task, suggesting that semantically similar categories within the same macro-domain remain difficult to discriminate.

Future models should address this in three directions:

1. Multi-label classification: Real posts frequently disclose multiple types of sensitive information simultaneously (e.g., health condition and geolocation and family relationship); a single dominant-label assumption loses this structure and underestimates cumulative impact.

2. Graded sensitivity : Instead of a binary sensitive/not label, models should predict a continuous or ordinal sensitivity score per category, enabling finer risk differentiation.
3. Uncertainty quantification : Predictive entropy or conformal prediction intervals allow the system to flag ambiguous cases for human review, activate human-in-the-loop workflows selectively, and calibrate decision thresholds according to the asymmetric cost of errors (false negatives on health or political content are typically more harmful than false positives).

Auxiliary tasks should be integrated as multi-task objectives to reduce the false-positive rate on adversarial examples that are the defining challenge of SPeDaC [27].

6.2.4 Generalization Across Platforms and Label Schemes

Empirical evidence from this thesis indicates that models trained on one platform (e.g., Twitter-like microblogging) do not transfer neatly to others (e.g., Facebook-like status updates or Reddit-like threaded discussions). This mismatch reflects differences in the linguistic register (short-form vs. long-form text, emoji density, hashtag morphology), discourse structure (flat timelines vs. nested threads) and annotation philosophies (perception-driven vs. legal/ontological vs. NER-derived). The NLP-PRISM survey systematically argues that cross-platform drift and representation leakage are persistent, under-addressed challenges: even when explicit identifiers are masked, latent demographic and stylistic signals remain exploitable, and membership/attribute inference attacks achieve strong performance (MIA AUC up to 0.81; AIA accuracy up to 0.75) across multiple tasks and model families [51].

To address this, future work should combine three complementary strategies.

- **Representation alignment** Lightweight domain adaptation (adversarial alignment, contrastive pretraining on mixed-domain corpora, adapter-tuning for multilingual LLMs) can reduce the distribution shift with-

out sacrificing the fine-grained sensitivity cues learned in-domain. Crucially, alignment must also bridge label-scheme heterogeneity: mapping perception-based binary labels (e.g., [33]) to ontology-based categories (e.g., SPeDaC [27]) requires explicit schema correspondences rather than ad-hoc relabeling.

- **Test-time and few-shot recalibration.** Real deployments rarely allow full retraining. Small calibration sets (dozens to a few hundred instances) can be used to retune thresholds, class priors, and confidence calibration curves on the fly, improving precision–recall trade-offs under new community norms (e.g., higher baseline disclosure of location terms in travel communities or higher prevalence of indirect self-reference in support groups).
- **Standardized out-of-domain protocols** Benchmarks should provide splits that simulate deployment (train on $A+B$, test on unseen C), report stability metrics (expected degradation, variance across seeds and domains), and include references to annotation philosophies. This discourages overfitting to dataset idiosyncrasies and rewards architectures that truly generalize.

6.2.5 Integrating Network Structure and Propagation Dynamics

Content sensitivity is a necessary but insufficient predictor of risk. Exposure depends on who speaks (centrality/tier), who listens (audience composition and community overlap), and how content travels (reshare/reply cascades, cross-community spillover). Alam aggregate graph structure (SimRank similarity, PageRank importance) with attributes and text into a composite CPRS score, demonstrating complementary predictive power from topology-aware features [50].

Advancing this line requires:

- audience-aware exposure modeling that approximates effective reach under privacy constraints (e.g., hashed/aggregated follower statistics, de-

gree/centrality summaries, community entropy as a dispersion proxy);

- detection of propagation anomalies (z-scores vs. user-specific baselines) to dynamically increase risk when diffusion departs from the user’s historical norm;
- analysis of amplification pathways (bridging nodes, assortative mixing) to estimate cross-community spillover before publication, enabling proactive warnings.

All computations should observe data-minimization and privacy-preserving analytics (on-device aggregation, federated analytics) to remain compliant with GDPR constraints.

6.2.6 Explainability and Actionable Mitigation

Risk scores should enable intervention, not only diagnosis. Without explanations and concrete guidance, users rarely adapt their behavior (privacy paradox) [16]. Future systems should, by design, produce the following:

- **Multi-level explanations:** token/span-level saliency (which phrases/entities drive sensitivity); feature-level decomposition (content vs. exposure contribution, with quantitative shares); decision-level rationales mapping to legally interpretable categories (e.g., GDPR Art. 9) and audience context.
- **Prepublication mitigations:** contextual paraphrasing to retain communicative intent while reducing inferability (e.g., abstracting precise conditions, obfuscating place/time granularity); visibility/timing adjustments and metadata stripping; unlinking of posts that are individually harmless but jointly revealing (sequence-aware linting).
- **What-if sandbox:** recompute risk after suggested edits, quantifying expected reduction so that users can choose the lowest-cost mitigation achieving a desired risk target.

Evidence from LLM studies indicates that post-hoc anonymization alone is insufficient against implicit inference; preventive, pre-publication edits deliver materially better protection [42].

6.2.7 Privacy-by-Design and Regulatory Alignment

Operational systems must implement privacy by design (GDPR Art. 25) and ensure legal interpretability. Practically, this implies:

- ephemeral processing of user text and storage of aggregate scores rather than raw content;
- audits for memorization and leakage (MIA/AIA) on models that ingest user data, as recommended by recent surveys and empirical evaluations [42], [51];
- fairness assessments across demographic and linguistic groups to avoid systematic over-penalization.

Taxonomies should explicitly map to GDPR/DVP (as in SPeDaC [27]), allowing the principled weighting of impact dimensions and facilitating by the oversight of data protection authorities. Where graph or audience data cannot be centralized, federated learning/analytics should be preferred, and only differentially private aggregates should be exported.

6.2.8 Future Privacy Benchmarks: Coverage, Structure, Interoperability, and LLM Evaluation

To support operational risk measurement, the benchmarks themselves must evolve along five axes.

Multilingual and cross-cultural coverage. Balance major language families, include code-mixed/dialectal varieties, recruit culturally competent anno-

tators, and report per-language/culture metrics. This prevents overfitting to Anglophone norms and exposes transfer gaps that matter in deployment.

Temporal and conversational modeling. Move beyond isolated instances to user timelines and threaded dialog; where feasible, include privacy-preserving cross-platform anchors. Annotate roles (self vs. third-party disclosure), thread outcomes (escalation, deletion), and design time-aware splits to evaluate cumulative risk and avoid temporal leakage.

Annotation interoperability via shared ontologies. Anchor labels in an actively maintained ontology (e.g., DPV/DPV-PD), provide dual-view annotations (ontological categories + perceived severity), and publish machine-readable schemas and mapping tables so models can train/evaluate across corpora without lossy relabeling.

Span-level granularity. Provide span/BIO annotations to localize sensitive content precisely, support multi-span, multi-label cases, and enable evaluation of targeted mitigations via what-if risk recomputation.

LLM-oriented protocols. Include tasks that probe explicit detection with localization, implicit inference from indirect cues, robustness to adversarial multi-turn elicitation, and mitigation efficacy. Report calibration, expected cost for asymmetric errors and leakage metrics (MIA/AIA), aligned with documented LLM risks [42], [51].

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